

A SIX-STATE MINIMAL TIME SOLUTION TO THE FIRING SQUAD SYNCHRONIZATION PROBLEM*

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Abstract. From Balzer's work (1967), we know that the firing squad synchronization problem has a minimal-time solution with eight states. We show that such a solution exists with only six states. Our method is somewhat different from all previous ones: the initial line is iteratively divided in two unequal parts so that each new right part can be treated as the homothetical image of a shorter initial line.

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1. Introduction

1.1. The firing squad synchronization problem was introduced by Moore in 1962 [3]. One considers a finite—but arbitrarily long—ordered line of n finite-state machines numbered from 1 to n . All machines numbered from 2 to $n - 1$ are identical and called soldiers; the machine numbered 1 is also called the *general* and the machine numbered n the *right-end soldier*. These n machines work synchronously; the state of a machine at time $t + 1$ depends only on the states at time t of itself and its (one or two) neighbours. For $t \leq 0$, all machines are in the same state, called the quiescent state. At $t = 1$, an external intervention forces the general into a new state all other machines still being in the quiescent state. Afterwards there is no external intervention, the line evolves as an isolated system.

* Advice to the reader: figures gain in readability when colouring the state-diagrams, one colour for each state.

The problem is to define finite sets of states and transition rules (for the three types of machines) with a distinguished state, called the 'fire state' so that, whatever the length of the line, all machines enter the fire state for the first time at the very same moment (called the synchronization time $t(n)$, which obviously depends on the length n of the line).

1.2. One can show that, necessarily, $t(n) \geq 2n - 1$. Intuitively, $2n - 1$ is the minimal time for the general to send a message to the right-end soldier and to get back an answer.

A minimal time solution of the synchronization problem is a family of finite sets of states and transition rules for which $t(n) = 2n - 1$. An N -state solution of the synchronization problem is a solution for which the union of the sets of states of the three types of machines is of cardinality at most N .

1.3. MacCarthy and Minsky proved the existence of solutions [5, pp. 28-29 and 282-283]; these solutions work in times $3n - 1$, $\frac{5}{2}n - 1, \dots$. A minimal time solution was presented by Waksman [4]; this solution uses sixteen states. Balzer presented an 8-state minimal time solution [1]. We present here a 6-state minimal time solution.

1.4. To study this problem, it is natural and usual to consider the set of pairs (K, t) with $1 \leq K \leq n$ and $1 \leq t \leq t(n)$. This plane set of pairs can be used in two ways.

(1) The state diagram is obtained by indicating the site values $\langle K, t \rangle$ (i.e., the state of machine K at time t). Graphically, we shall represent the site (K, t) by a unit square and get an N -coloured tiling of the rectangle formed by these $n \times t(n)$ unit squares.

(2) Geometrical diagrams are obtained by indicating the action of some distinguished transition rules which correspond intuitively to the propagation of signals. Graphically, this gives continuous lines through the portion of plane $[1, n] \times [1, t(n)]$.

1.5. To avoid the consideration of three cases of transition rules corresponding to the three kinds of machines, it is convenient to introduce

- a new artificial state denoted X ;
- two artificial machines always in state X , delimiting the line, having ranks 0 and $n + 1$.

This trick allows us to represent the transition rules of an N -state solution by a family of $N - 1$ matrices of states with N rows and N columns as follows:

- we denote by q_0 the fire state, by q_1 the state X , and by $q_2, \dots, q_N$ the remaining states;
- if at time t , the machines numbered $K - 1, K, K + 1$ (where $K \in \{1, \dots, n\}$) are in respective states q_a, q_b, q_c (where a, b, c are in $\{1, \dots, N\}$), then at time $t + 1$, the machine K is in state q where q is the element of the b th matrix on row a and column b .

Remark. (1) Since all machines have to enter the fire state at the very same time, we do not consider transition rules where one of the a , b , c is 0.

(2) The elements of these matrices are among $q_0, q_2, q_3, \dots, q_N$.

(3) Thus, all machines $1, 2, \dots, n-1, n$ are considered as identical: they have the same set of internal states $\{q_0, q_1, q_2, q_3, \dots, q_N\}$ and the same transition rules.

The way machines 1 and n , the general and the right-end soldier, can be distinguished is as follows: machine 1 (respectively machine n) is the only one to have a left (respectively right) neighbour in state q_1 and so to make use of the related transition rules indicated in the first row (respectively column) of each matrix.

1.6. The idea of the earlier solutions (MacCarthy and Minsky) is the following:

- after the external intervention the general generates two waves which propagate through the line at different speeds (cf. Fig. 1);
- the fast wave is reflected by the right-end soldier;
- this reflection meets the slow wave at the middle of the line;
- in this way, the initial line is broken into two new lines having equal length, which evolve independently (depending on the parity of n , these two lines are disjoint or have a common element);
- the right line evolves in a way 'homothetical' to that of the initial line so that its general is its leftmost machine and the left line evolves in a way 'symmetric' and 'homothetical' to that of the initial line so that its general is its rightmost machine;
- this dichotomy is iterated until lines with length two are obtained: then the fire state appears.

1.7. The common features of Waksman's and Balzer's solutions [4, 1] are the following:

- after the external intervention, the line generates a family of waves, all of which seem to come from the general;
- the two fastest ones ($G_1G_2G_3$ and G_1G_3 in fig. 2(a)) act as in Minsky and MacCarthy's solutions and break the initial line at G_3 creating two new lines having equal length;
- the right line (consisting of machines numbered from $\frac{1}{2}(n+1)$ to n), which is created at time $\frac{3}{2}n-1$, has in fact begun its evolution at time n , its general being the machine n ; it evolves in a way such that the trapezoid of sites $F_3G_3G_2F_2$ is symmetric and homothetical to the trapezoid of sites $F_2G_2G_1F_1$ associated to the initial line;
- the remaining portion of the initial line (machines 1 to $\frac{1}{2}(n+1)-1$) is also iteratively broken by the meeting of G_2F_1 with slower and slower waves all seemingly starting at site G_1 .

This remaining portion evolves in such a way that the triangle of sites $F_1G_3F_3$ is 'homothetical' to the triangle of sites $F_1G_2F_2$. The two triangles of sites $F_1G_3F_3$ and $F_2G_3F_3$ are symmetric so that the two machines 1 and n (the first and second generals) are synchronized.

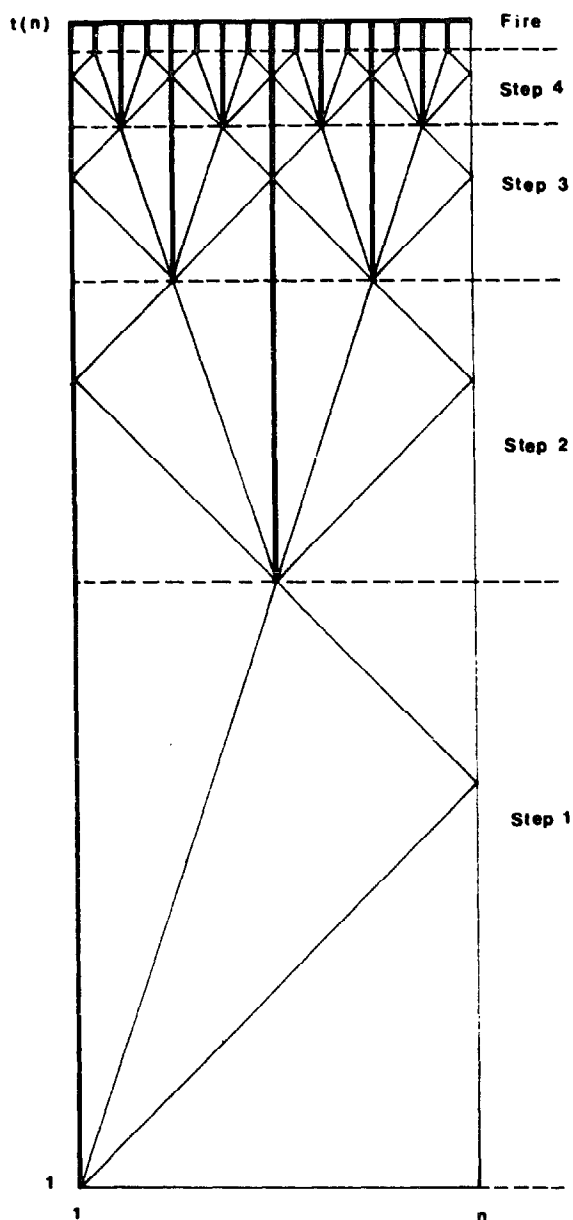


Fig. 1.

In case n is odd, Fig. 2(a) shows the first step of this iterative process. In case n is even, Fig. 2(b) shows the needed (easy) modifications. Figure 2(c) shows the iterative process in the idealized (though impossible) case where all breaks are as in Fig. 2(a).

1.8. The previous solutions seem symmetric since the initial line is broken into equal parts. However the general is alternatively at the right end or at the left end of the successive lines. This fact induces an irregular character and introduces superfluous internal states. We shall give a minimal time solution in which the general is always the left-end soldier.

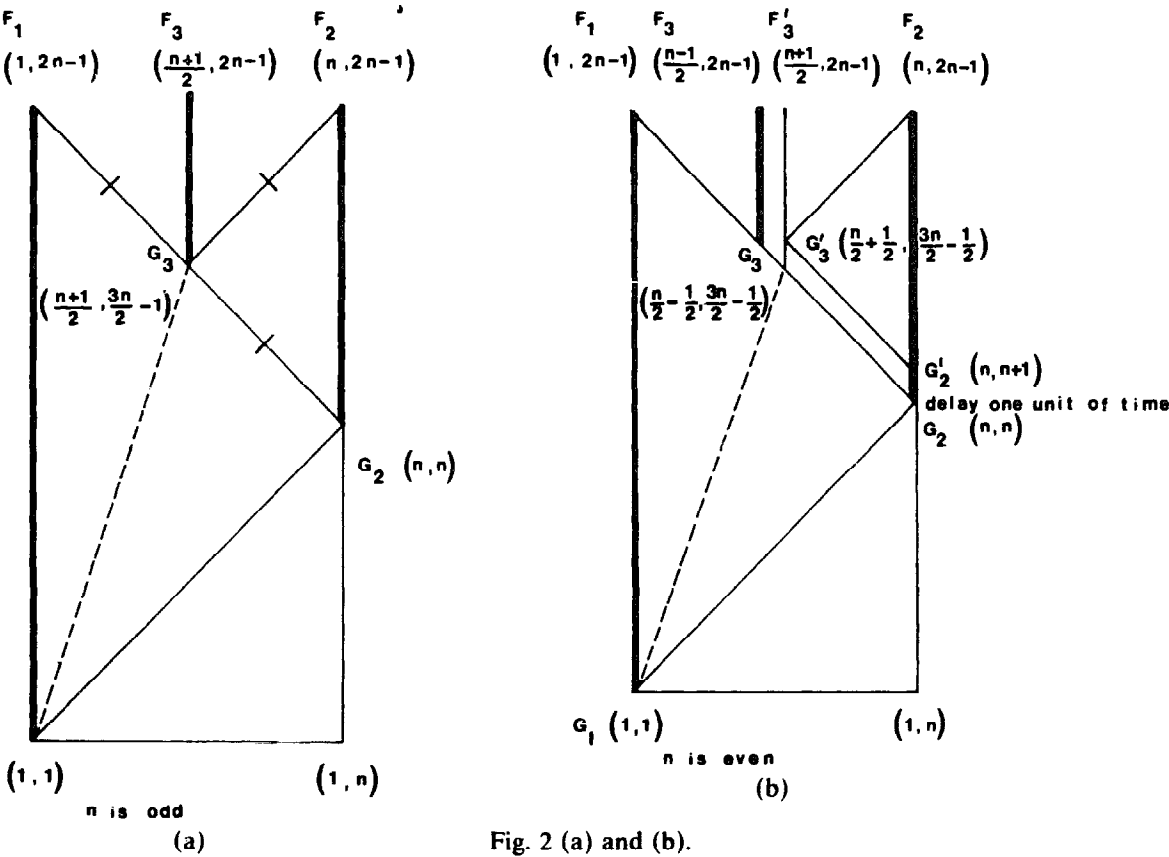


Fig. 2 (a) and (b).

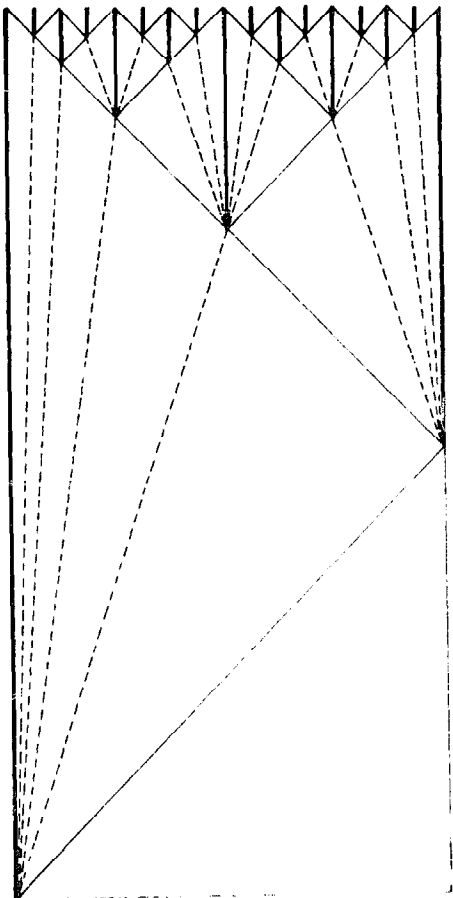


Fig. 2(c).

2. Geometry and delays

2.1. As indicated above, in the solution presented here, all created lines will have as a general their leftmost machine. This can be done in a very convenient way of breaking lines at $\frac{2}{3}$ of their length instead of at $\frac{1}{2}$.

We shall first describe our solution with continuous geometrical diagrams: such a description applies to an idealized case where all intersections of lines have integral coordinates (in fact this ideal case does not exist!).

In this situation we have the following property shown by Fig. 3(a). The initial line is broken at site G_2 (so that $G_2F_1 = 2R_1G_2$). The new line consists of machines $\frac{2}{3}n, \dots, n$. The wave G_2F_1 (which is the progression from site G_2 of the reflection of the initial signal G_1R_1) has the same length as the wave $G_2R_2F_2$ (which is the progression of the wave created in the new line on site G_2 , reflected on site R_2 and going back on F_2 to the general of the new line).

In particular, these waves G_2F_1 and $G_2R_2F_2$ reach their respective generals at the very same time, thus synchronizing the two generals, i.e., machines 1 and $\frac{2}{3}n$. This is the basic step of an iterative process which leads to the synchronization of the whole line.

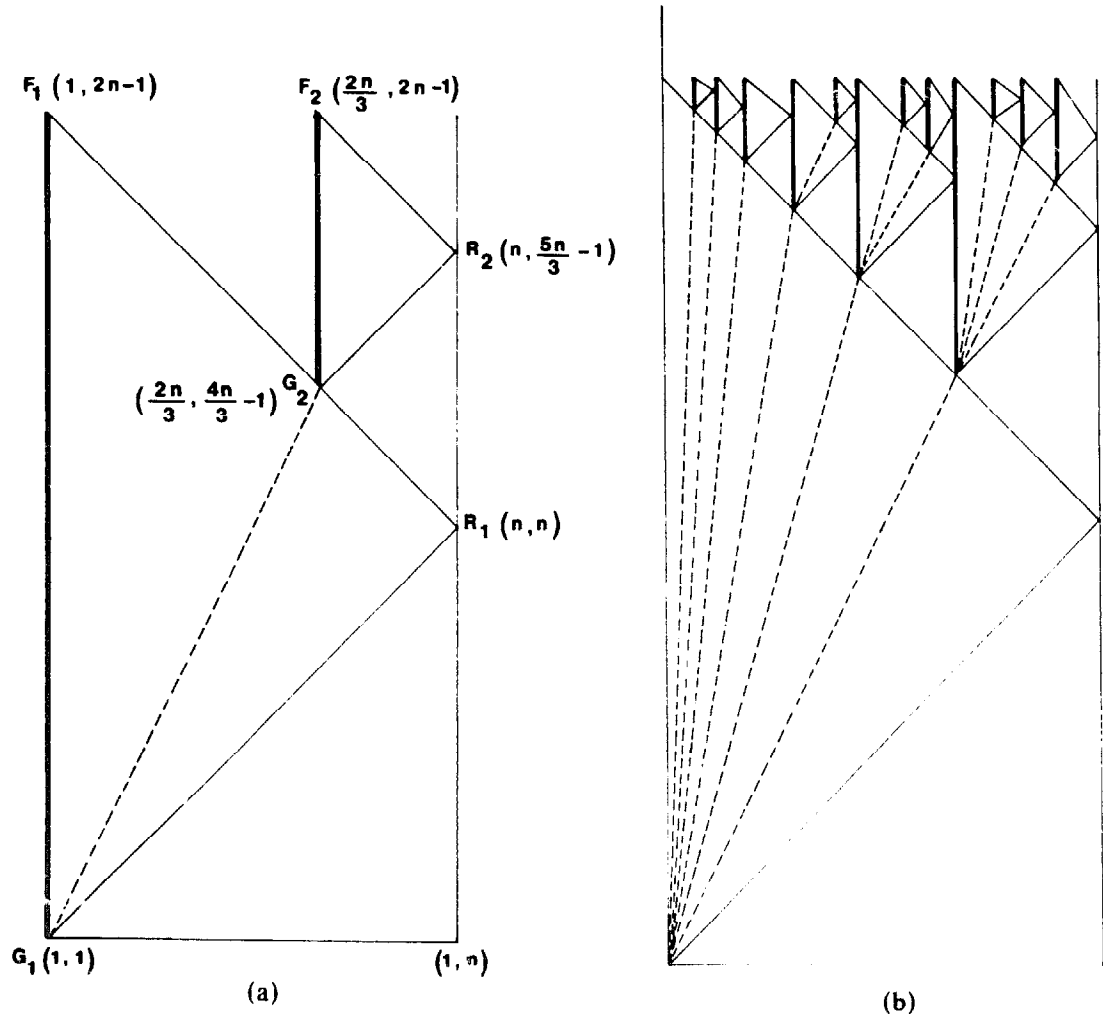


Fig. 3.

Remark. This situation only occurs if the breaking point is on machine $\frac{2}{3}n$.

As in Waksman's and Balzer's solutions [4, 1], the created line and the remaining portion of the initial line (which consists of machines $1, \dots, \frac{2}{3}n - 1$) evolve independently from time $\frac{4}{3}n - 1$ to time $2n - 1$:

- the evolution of the new line from time $\frac{4}{3}n - 1$ up to time $2n - 1$ is homothetical to that of the initial line from time 1 up to time $2n - 1$;
- the remaining portion of the initial line will also be broken at $\frac{2}{3}$, i.e., at machine $(\frac{2}{3})^2n$, creating a second new line consisting of machines numbered from $(\frac{2}{3})^2n$ to $\frac{2}{3}n - 1$, and a second remaining portion of the initial line;
- this new left portion is also broken and so on.

In this way an iterative process is set up. The creation of the new line with general $\frac{2}{3}n$ at site G_2 is the result of the meeting of the reflection R_1F_1 of the initial wave with the slow wave G_1G_2 .

The remaining portion of the initial line (machines 1 to $\frac{2}{3}n - 1$) is also iteratively broken by the meeting of R_1F_1 with slower and slower waves all (seemingly) starting at site G_1 .

Figure 3(b) shows this iterative process (with all waves) in the idealized case.

2.2. In Figs. 3(a) and (b) we have supposed that all sites have integral coordinates. In the discrete situation where n is an integer, we have to modify the basic step (Fig. 3(a)) according to the 'ternarity' (the remainder modulo 3) of n .

In the sequel, we shall no longer consider geometrical diagrams but state-diagrams.

The initial wave fills sites (l, l) and its reflection fills sites $(l, 2n - l)$ (for $l \in \{1, \dots, n\}$). Suppose that the new general is created at site $G_2 = (K, 2n - K)$. The reflection of the initial wave reaches the old general (machine 1) at time $(2n - K) + (K - 1) = 2n - 1$.

If the new general (machine K) becomes active after a delay of j units of time, then

- the new initial wave starts at site $(K, 2n - K + j)$,
- it reaches the right-end soldier (machine n) at site $(n, 2n - K + j + n - K)$,
- and goes back to the new general at time $2n - K + j + (n - K) + (n - K) = 4n - 3K + j$. In this case, both the new and old generals (machines K and 1) will be synchronized if $2n - 1 = 4n - 3K + j$, i.e., $3K = 2n + j + 1$. This equation in K and j is solvable with the constraint that $j \in \{0, 1, 2\}$.

Suppose that the initial line has length $n = 3p + i$ with $i \in \{1, 2, 3\}$ and $p \geq 1$; then,

- if $i = 1$, we get $K = 2p + 1$ and $j = 0$;
- if $i = 2$, we get $K = 2p + 2$ and $j = 1$;
- if $i = 3$, we get $K = 2p + 3$ and $j = 2$.

The value of j will be called the *delay* for the activation of the new general. In all three cases, the new line consists of $p + 1$ machines numbered from $K = 2p + i$ to $n = 3p + i$ and the delay is $j = i - 1$. Figure 4 shows these three different cases.

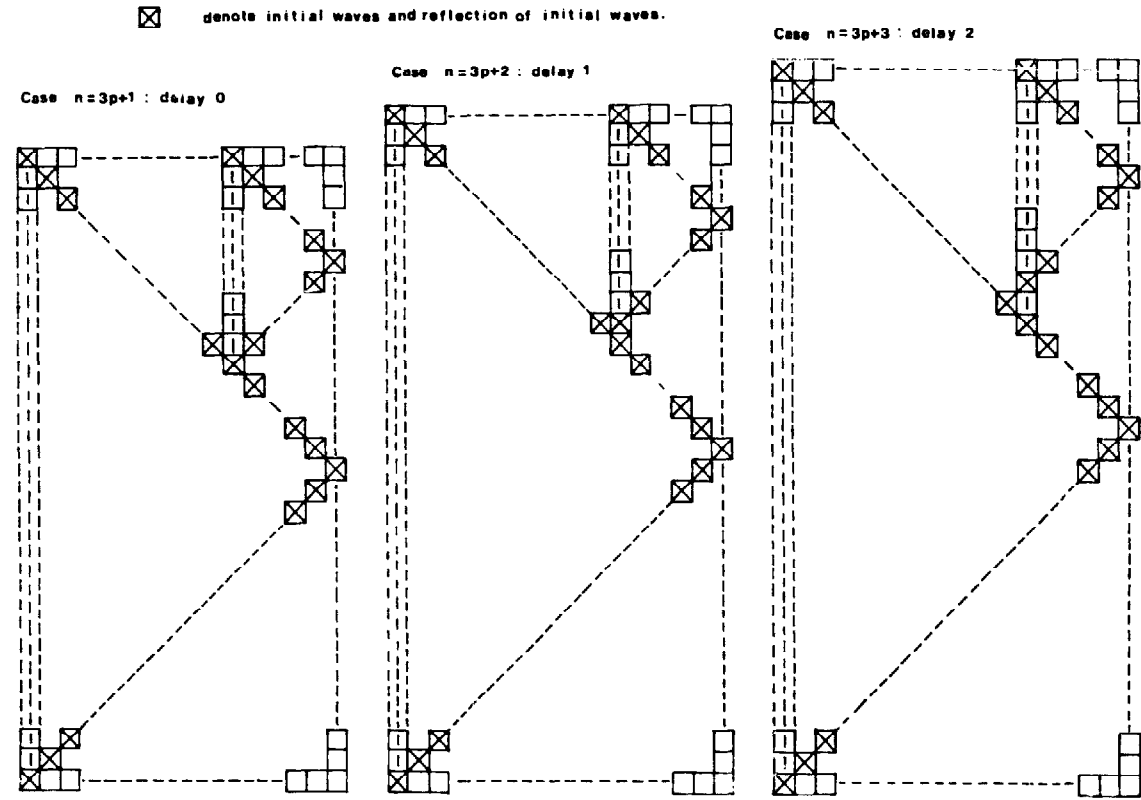


Fig. 4.

Remark. The slow wave progresses at speed $\frac{1}{2}$ (so that it is graphically represented by a 'line of sites' of slope 2). Machine K receives this slow wave at time $2K - 1$ and the reflection of the initial wave at time $n + n - K = 2n - K$. The value of K determined above is also the value for which

- the reflection of the initial wave arrives at machine K before (or at the same time as) the slow wave;
- the waiting delay between these two waves is minimal.

3. The scheme

We now study the generation of the family of slow waves. In fact, it is more convenient to introduce the waves on which successive right-end soldiers are created (rather than those corresponding to new generals).

3.1. Define the set of sites $\tilde{S}_1$ as the set of sites (K, t) such that, for some value of n ($n \geq 4$), the first break of a line of n machines occurs on site $(K + 1, t)$ so that machine K is the right-end soldier at site (K, t) (and the reflection of the initial wave is on machine K at time $t + 1$).

As seen in paragraph 2.2, if $n = 3p + i$, then machine $2p + i$ becomes a general at time $2n - (2p + i) = 4p + i$ so that

$$\tilde{S}_1 = \{(2p + i - 1, 4p + i) \mid p \geq 1; i \in \{1, 2, 3\}\}.$$

Observe that two sites (K, t) and (K', t') are 'related' if $K = K'$ and $|t - t'| = 1$, or if $|K - K'| = 1$ and $t = t'$. It is convenient to connect the set $\tilde{S}_1$ and introduce the set S_1 :

$$S_1 = \{(2p + i - 1, 4p + i - 1) | p \geq 1, i \in \{1, 2, 3\}\} \cup \tilde{S}_1.$$

Similarly, we define, for every positive i , the set of sites $\tilde{S}_i$ as the set of sites (K, t) such that, for some value of n , the i th break of the initial line of n machines occurs on site $(K + 1, t)$ (all these breaks are relative to the leftmost remaining portion of the initial line); and for $K \geq 2$ also S_i is defined by

$$S_i = \tilde{S}_i \cup \{(K, t - 1) | (K, t) \in \tilde{S}_i\}.$$

Observe that if machine K transmits the initial wave at time t , then it receives its reflection at time $t + 2(n - K)$ which has the same parity as t so that one unit of time out of two can be the moment of the arrival of its reflection. This phenomenon will be called the 'internal clock'. It allows us to use half of the time to obtain the synchronization, either with particular constraints imposed (cf. [3]) or with fewer states.

3.2. A 'connected wave' is by definition a nonempty set W of sites such that if $(K, t) \in W$, then either $(K + 1, t) \in W$ or $(K, t + 1) \in W$ and $(K - 1, t)$ and $(K + 1, t)$ are not both in W .

We define the *starting machine* $SM(W)$ and the *starting time* $ST(W)$ of W as follows: $SM(W)$ is the smallest K such that (K, t) is in W for some t and $ST(W)$ the smallest t such that $(SM(W), t)$ is in W .

To each site (K, t) with $K \geq 4$, we associate the site $NR(K, t)$ (NR for new right-end soldier) so that $NR(K, t)$ is the *new right-end soldier* after the first break in the evolution of the line of machines 1 to K , where the external intervention occurs at time $t - K + 2$ (so that the reflection occurs at time $t + 1$: as the initial wave reaches the right-end soldier on site (K, t) , the reflection occurs at time $t + 1$).

The domain of NR is $D = \{(K, t) | K \geq 4\}$ and, as seen in paragraph 2.2, if (K, t) in D is of the form $(K, t) = (3p + i, t)$ where $i \in \{1, 2, 3\}$ and $p \geq 1$, then

$$NR(3p + i, t) = (2p + i - 1, p + t + 1). \quad (1)$$

We observe that if W is a connected wave, then $W \cap D$ and its image $NR(W \cap D)$ are empty or are also connected waves. In fact, (1) shows (2): if $NR(K, t) = (K', t')$, then

$$\begin{aligned} NR(K, t + 1) &= (K', t' + 1) \quad \text{and} \\ NR(K + 1, t) &= \begin{cases} (K', t' + 1) & \text{if } K \equiv 0 \pmod{3}, \\ (K' + 1, t') & \text{if } K \not\equiv 0 \pmod{3}. \end{cases} \end{aligned} \quad (2)$$

If the starting machine of $W \cap D$ is of the form $SM(W \cap D) = 3p + i$ with $p \geq 1$ and $i \in \{1, 2, 3\}$, then $NR(W \cap D)$ is nonempty and its starting machine is $2p + i - 1$.

We now describe a process to obtain the connected wave $NR(W \cap D)$ from $W \cap D$ (cf. Fig. 5). This process is twofold. First we initialize the wave $NR(W \cap D)$, then we construct the whole wave from its starting site.

The initialization of $NR(W \cap D)$ cannot be done in a general setting: there is no way to obtain quickly the starting machine of $NR(W \cap D)$ from that of $W \cap D$. We suppose that $SM(W \cap D) = 3p_0 + i_0$ where $p_0 \geq 1$ and $i_0 \in \{1, 2, 3\}$, whence (using

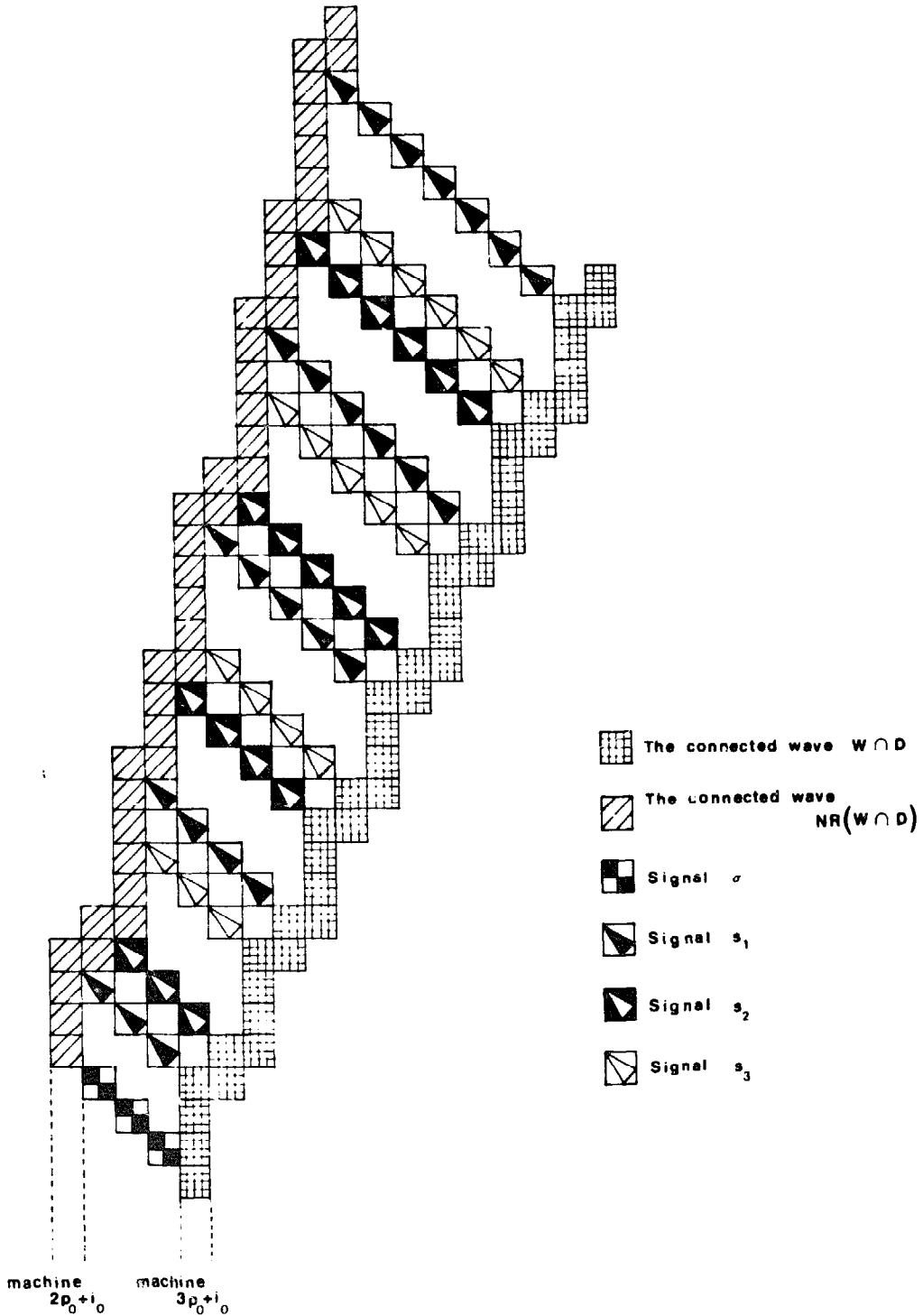


Fig. 5.

(1)) $SM(NR(W \cap D)) = 2p_0 + i_0 - 1$. This starting machine being fixed, we do construct the starting time of $NR(W \cap D)$ from that of $W \cap D$ by the following elementary process:

- from site $(SM(W \cap D), ST(W \cap D))$ a signal σ is emitted which propagates along the diagonal

$$\Delta(SM(W \cap D), ST(W \cap D)) \\ = \{(SM(W \cap D) - l, ST(W \cap D) + l) \mid 0 \leq l \leq SM(W \cap D) - 1\}.$$

- The starting time of $NR(W \cap D)$ is the moment when this signal σ reaches machine $SM(NR(W \cap D))$.

We now describe the construction of the wave $NR(W \cap D)$ from its starting site. First we observe, using equation (1) that both sites (K, t) and $NR(K, t)$ (for $K \geq 4$) belong to the same diagonal:

$$\Delta(K, t) = \{(K - l, t + l) \mid 0 \leq l \leq K - 1\}.$$

The content of (2) can be rephrased as follows:

- when one moves vertically on $W \cap D$ (i.e., from (K, t) to $(K, t+1)$), then the corresponding move by NR on $NR(W \cap D)$ is also vertical;
 - when one moves horizontally on $W \cap D$ (i.e., from $(3p+i, t)$ to $(3p+i+1, t)$), then the corresponding move by NR on $NR(W \cap D)$ is in two cases out of three also horizontal (cases $i=1$ or $i=2$), and in one case out of three vertical (case $i=3$).

Similarly to Balzer [1], we introduce three kinds of distinguished states s_1, s_2, s_3 propagating along the diagonals $\Delta(K, t)$ for those (K, t) in $W \cap D$ such that $(K+1, t)$ is also in $W \cap D$. Let $r \in \{1, 2, 3\}$ be such that $SM(W \cap D) \equiv r \pmod{3}$. We let the first such signal (the one which starts from a site $(SM(W \cap D), t)$) be s_r and we let these signals appear successively in the order $\dots s_1 s_2 s_3 s_1 \dots$. Thus, the only possible signal starting from machine K (if there is some) is s_j where

$$j \equiv r + (K - SM(W \cap D)) \pmod{3} \equiv K \pmod{3}. \quad (3)$$

Statement (2) can now be restated as follows (4): if $(K', t') = NR(K, t)$ where (K, t) is in $W \cap D$, then

- if (K', t') does not belong to any of these signals s_j (i.e., $(K, t+1) \in W \cap D$ and $(K+1, t) \notin W \cap D$), then $(K', t'+1) \in NR(W \cap D)$;
 - if (K', t') does belong to such a signal s_j where $j \equiv 1$ or $j \equiv 2 \pmod{3}$ (case $K \equiv 1$ or $K \equiv 2$ of (3)), then $(K'+1, t')$ is in $NR(W \cap D)$;
 - if (K', t') does belong to such a signal s_j where $j \equiv 0 \pmod{3}$ (case $K \equiv 3$ of (3)), then $(K', t'+1)$ is in $NR(W \cap D)$.

Figure 5 shows such a construction of $NR(W \cap D)$ from $W \cap D$.

3.3. As seen in paragraph 2.2, if a line has $3p+i$ machines, $i \in \{1, 2, 3\}$, then the new general which will be created will become active after a delay of $i-1$ units of time. We have noticed in paragraph 3.2 that the only possible signal starting from machine

$3p + i$ is s_i . This shows that s_i conveys the information: “If you become general, then be active after a delay of $i - 1$ units of time”.

3.4. Let S_0 be $\{(K, K - 1), (K, K - 2) \text{ for } K \geq 4\}$. The very definitions of S_1 and NR show that $S_1 = \text{NR}(S_0)$ (observe that $S_0 \subset D$, the domain of NR). Also, for all $i \geq 1$, we have $S_{i+1} = \text{NR}(S_i \cap D)$. Figure 6 illustrates these processes which give S_1 from S_0 and iteratively $S_2, S_3, \dots$ from $S_1, S_2, \dots$.

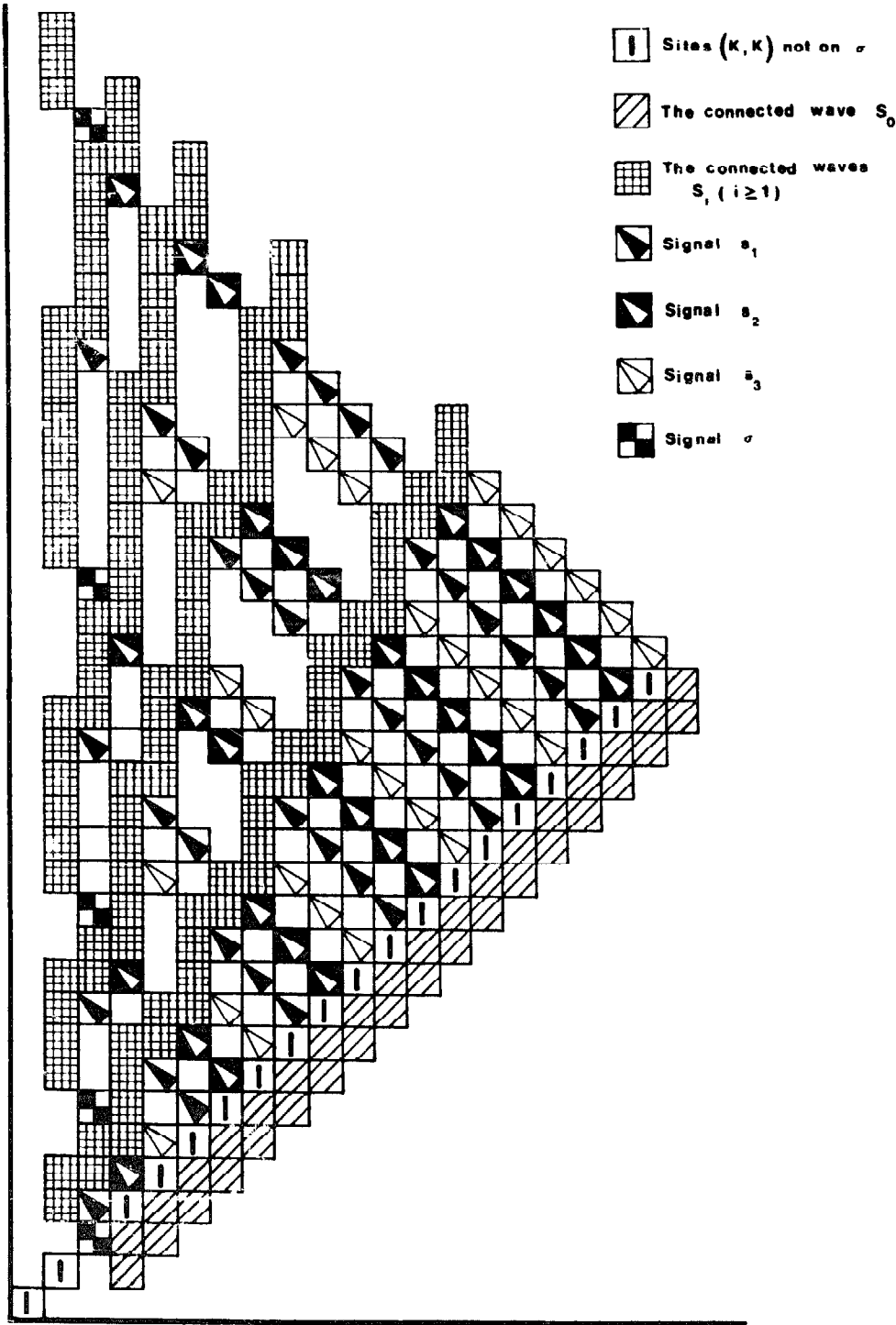


Fig. 6.

4. States and transition rules for an 8-state solution

We suggest the reader to constantly refer to Fig. 12 (which gives an instance of the synchronization process) as an illustration of the material developed in this section.

Besides the matrix notation for transition rules described in paragraph 1.5, it is convenient to denote by $(U, V, W) \rightarrow T$ the transition rule which asserts that if machines $K-1, K, K+1$ are at time t in states U, V and W , then, at time $t+1$, machine K is in state T .

4.1. The very statement of the synchronization problem introduces

- two particular states: the quiescent state (denoted L), and the fire (denoted F);
- obviously related transition rules $(L, L, L) \rightarrow L$ and $(L, L, X) \rightarrow L$ and $(X, L, L) \rightarrow L$. Clearly, for $t < K$, site (K, t) is in state L.

We now introduce states and transition rules convenient to set up the process described in Section 3: first we shall introduce such states in a loose way, then we shall severely reduce the set of states.

Note. It is essential to observe that the different notions associated to signals and waves, need not be characterized by particular states, but rather by particular situations, that is, triples of states (corresponding to machines $K-1, K, K+1$).

4.2. We present a first tentative set of states and transition rules translating the inductive step of the construction of the S_i 's via the signals s_j 's (cf. Section 3). We introduce four particular states A, B, C, S:

- all sites in $\bigcup_{i \geq 1} S_i$ will be in state S;
- all sites 'between' S_i and S_{i+1} will be in state A, B, or C. States A, B, and C will be distributed so that
- signal s_1 (respectively, s_2, s_3) is on site (K, t) iff (K, t) is in state A (respectively, B, C) and $(K+1, t)$ is in state B (respectively, C, A).
- if s_j and $s_{j'}$ are successive signals going from S_i to S_{i+1} , then all sites 'between' s_j and $s_{j'}$ (and S_i and S_{i+1}) are in the same state: the one common to all sites in $S_{j'}$.

Following the different elements of the construction in Section 3, we now enumerate, comment and represent adequate transition rules in Fig. 7 (anticipating paragraph 4.3, state S is indicated by L in Fig. 7).

(1) S_i emits s_j : We get three rules:

$$(A, S, S) \rightarrow B, \quad (B, S, S) \rightarrow C, \quad (C, S, S) \rightarrow A.$$

The first one can be interpreted as follows: at time t , sites (K, t) and $(K+1, t)$ are in S_i . Site $(K-1, t)$ is in state A; this means that the last signal emitted by S_i was s_3 , coded by $(. C A)$. Site (K, t) has to emit signal s_1 : in order that site $(K-1, t+1)$ be in signal s_1 , site $(K, t+1)$ must be in state B (and site $(K-1, t+1)$ must be in state A since s_1 is coded by $(. A B)$).

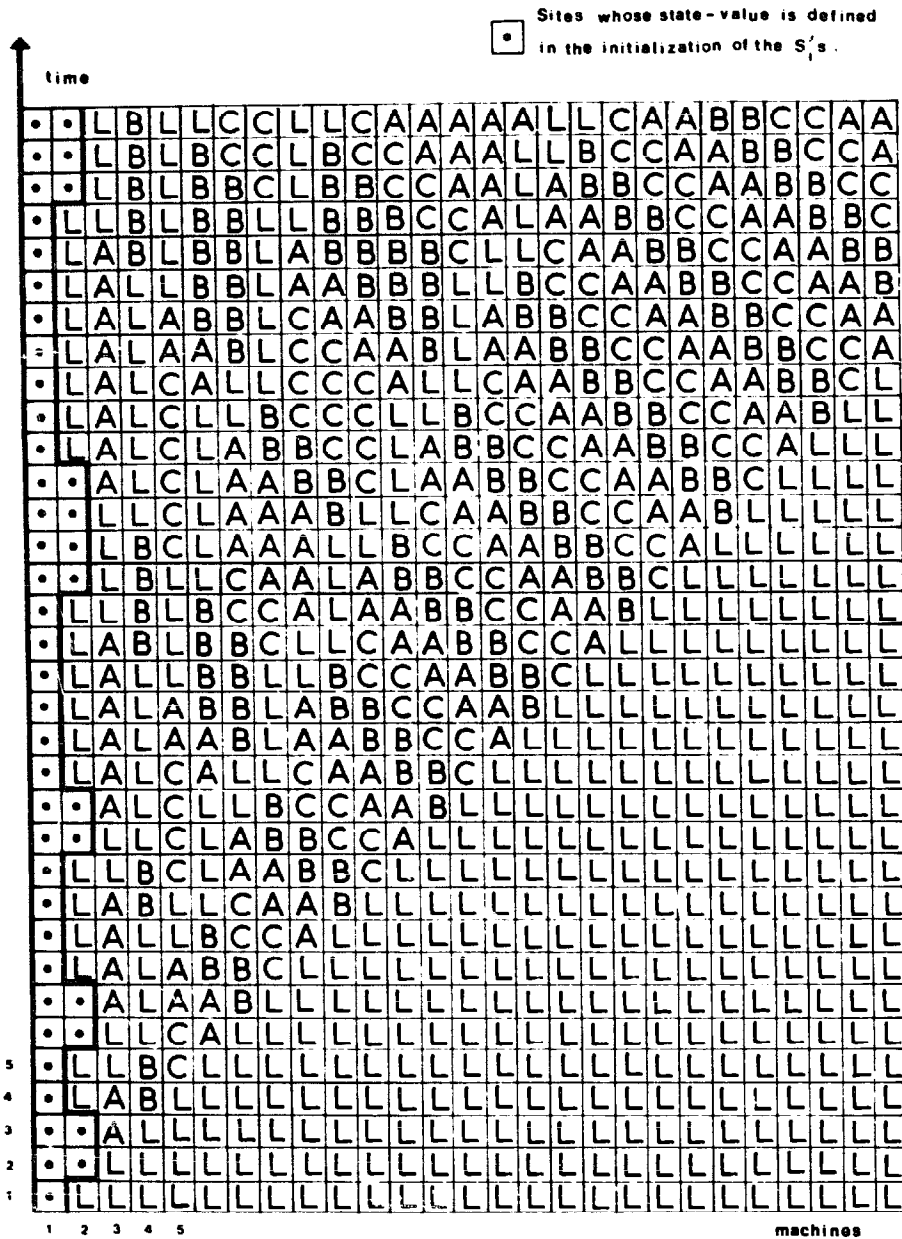


Fig. 7.

(2) S_i does not emit any s_j : We get nine rules:

$$(A, A, S) \rightarrow A, \quad (B, B, S) \rightarrow B, \quad (C, C, S) \rightarrow C,$$

$$(C, A, S) \rightarrow A, \quad (A, B, S) \rightarrow B, \quad (B, C, S) \rightarrow C,$$

$$(S, A, S) \rightarrow A, \quad (S, B, S) \rightarrow B, \quad (S, C, S) \rightarrow C.$$

Rules $(\cdot, A, S) \rightarrow A$ start the filling of a new diagonal between two successive signals s_3 and s_1 (with the three possible left neighbours). Other rules have similar meaning.

(3) *Filling diagonals*: We get two sets of rules:

(i) $(A, A, B) \rightarrow B, \quad (B, B, C) \rightarrow C, \quad (C, C, A) \rightarrow A,$

$$(ii) \quad \begin{cases} (A, B, B) \rightarrow B, & (B, C, C) \rightarrow C, & (C, A, A) \rightarrow A, \\ (B, B, B) \rightarrow B, & (C, C, C) \rightarrow C, & (A, A, A) \rightarrow A. \end{cases}$$

Rule $(A, A, B) \rightarrow B$ allows the propagation of signal s_1 : if (K, t) is on signal s_1 , then $(K-1, t+1)$ is also on signal s_1 so that $(K, t+1)$ has to be in state B (and $(K-1, t+1)$ in state A).

Rules $(A, B, B) \rightarrow B$ and $(B, B, B) \rightarrow B$ do insure the filling of diagonals between successive signals s_j .

(4) S_{i+1} receives s_j : According to statement (4) in paragraph 3.2, we get three rules:

$$(S, A, B) \rightarrow S, \quad (S, B, C) \rightarrow S, \quad (S, C, A) \rightarrow A.$$

Rule $(S, A, B) \rightarrow S$ can be interpreted as follows: site (K, t) is on signal s_1 and has its left neighbour in S_{i+1} . This means that S_{i+1} receives a signal s_1 on site $(K-1, t+1)$ and has to 'move right': site $(K, t+1)$ has to be in S_{i+1} , hence in state S.

Rule $(S, B, C) \rightarrow S$ is similar to signal s_2 . Rule $(S, C, A) \rightarrow A$, interpreted in the same way, means that S_{i+1} receives a signal s_3 on site $(K-1, t+1)$, hence has to 'move vertically', so that site $(K, t+1)$ is also outside S_{i+1} . Since the next signal will be s_1 , site $(K, t+1)$ has to be in state A.

(5) S_{i+1} does not receive any s_j : We get three rules:

$$(S, A, A) \rightarrow A, \quad (S, B, B) \rightarrow B, \quad (S, C, C) \rightarrow C.$$

These rules complete the filling of diagonals between successive signals.

(6) *Vertical moving of S_i and S_{i+1}* : We know that if (K, t) is on S_i and $(K+1, t)$ is not on S_i , then $(K, t+1)$ has to be on S_i . This leads to eleven rules:

$$(S, S, V) \rightarrow S, \quad (U, S, W) \rightarrow S,$$

where U and W are A, B or C and V is B or C.

Observe that any environment SSV corresponds to a horizontal move of some S_i , hence to the reception of signals s_1 or s_2 (and not s_3). The middle S in SSV must come from A or B (and not C) with right neighbours BB or CC (recall that signals s_1 and s_2 are coded by AB and BC). Thus, V is obtained as the result of the rules for environments ABB and BCC: V is B or C (and not A).

(7) *Construction of S_1 from S_0* : We now attribute states A, B or C to sites (K, K) for $K \geq 3$. If $K \equiv 1 \pmod 3$, then site $(K, K-1)$, which is on S_0 , emits a signal s_1 . Since s_1 is coded by AB and $(K-1, K)$ is on s_1 , sites $(K-1, K)$ and (K, K) have to be in states A, B. So we put state B on site (K, K) . Similarly, if $K \equiv 2 \pmod 3$ (respectively $K \equiv 3 \pmod 3$), then site (K, K) will have state C (respectively A). This gives nine transition rules:

$$\begin{aligned} (A, L, L) &\rightarrow B, & (B, L, L) &\rightarrow C, & (C, L, L) &\rightarrow A, \\ (A, B, L) &\rightarrow B, & (B, C, L) &\rightarrow C, & (C, A, L) &\rightarrow A, \\ (S, A, L) &\rightarrow A, & (S, B, L) &\rightarrow B, & (S, C, L) &\rightarrow C \end{aligned}$$

(the last three are in fact useless but not obviously so).

Observe that if (K, K) has state A (respectively B, C), then it does know: “if the reflection occurs at the next unit of time, then the line has $K+1$ machines and $K+1 \equiv 1 \pmod 3$ (respectively, 2, 3) so that the new general will become active after delay 0 (respectively 1, 2)”.

4.3. We do observe the following key fact: *replacing S by L in all previous rules (1) to (7), one gets a compatible set of rules.* This permits to eliminate state S in profit of L.

4.4. We shall now continue our comments on Fig. 7.

(8) *Initialization of the construction of the S_i 's ($i \geq 1$):* According to paragraph 3.2, we observe that the starting machine of S_{i+1} ($i \geq 0$) is machine 2, and the starting site of S_{i+1} is obtained via the signal σ emitted by machine 4 at the smallest time t for which (K, t) is in S_i (so that we also have $(3, t)$ in S_i). Observe that $t = ST(S_i \cap D)$ and $ST(S_{i+1}) = ST(S_i \cap D) + 2$ (it takes two units of time for signal σ to go from machine 4 to machine 2 along a diagonal).

We introduce two states G and H:

- G marks all sites $(1, t)$ for $t \geq 1$;
- H marks all sites $(2, t)$ except those already marked by L (i.e., those in $\bigcup_{i \geq 0} S_i$).

In particular, since sites $(3, ST(S_i \cap D))$ and $(4, ST(S_i \cap D))$ both have state L, site $(2, ST(S_i \cap D))$ has to be in state H. Also, since σ reaches machine 2 at time $ST(S_i \cap D) + 2$, site $(2, ST(S_i \cap D) + 1)$ is not in S_{i+1} , hence has to be in state H.

Apart from σ , the first signal emitted by S_i ($i \geq 0$) starts from machine 4 at time θ such that $(4, \theta)$ and $(5, \theta)$ are in S_i . This signal is s_1 since $4 \equiv 1 \pmod 3$. Since s_1 is denoted by states A, B on adjacent sites, sites $(3, \theta + 1)$ and $(4, \theta + 1)$ are in states A and B.

In order not to introduce new transition rules we attribute state A to all sites $(3, t')$ for $1 + ST(S_i \cap D) \leq t' \leq \theta$.

Now, to all sites below the first signal s_1 emitted by S_i , a state has been attributed (cf. Fig. 8). Note that signal σ is coded by the environment HAL.

This whole process necessitates the introduction of the following rules:

$(X, G, L) \rightarrow G, \quad (X, G, H) \rightarrow G, \quad \text{Persistence of G;}$

$(G, L, L) \rightarrow H, \quad (G, H, L) \rightarrow H, \quad \text{Introduction and (partial persistency) of H;}$

$(G, H, A) \rightarrow L, \quad \text{machine 3 is always in state A or L, when A comes, it carries signal } \sigma \text{ so that machine 2 gets into } S_{i+1} \text{ and has state L;}$

$(G, L, A) \rightarrow L, \quad \text{(partial) persistency of L on machine 2;}$

$(H, L, L) \rightarrow A, \quad (H, A, L) \rightarrow A, \quad \text{machine 3 emits signal } \sigma;$

$(H, L, B) \rightarrow L, \quad \text{(partial persistency of L on machine 3, machine 4 is always in state A, B or L).}$

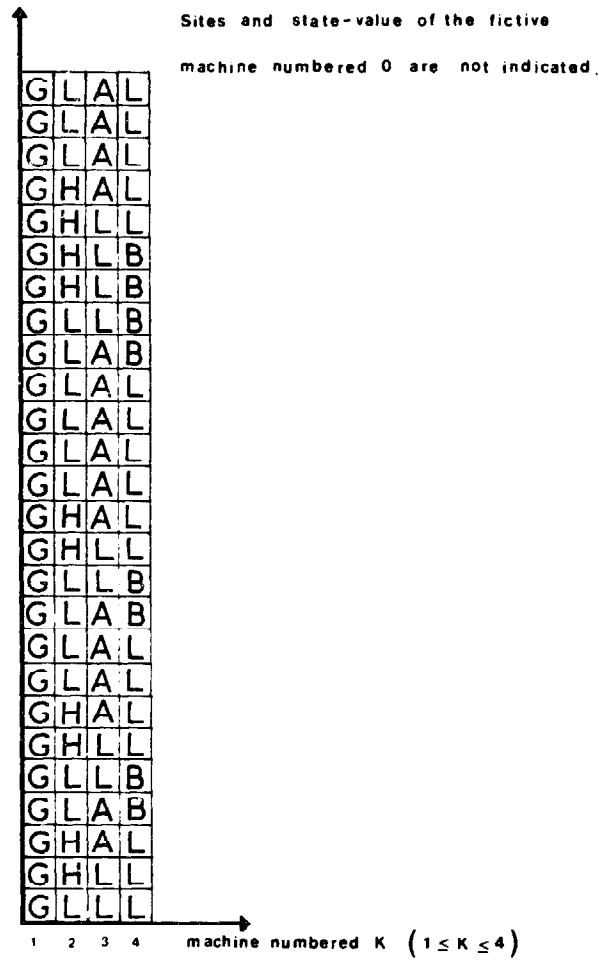


Fig. 8.

We remark that transition rules (1)–(8) (where L replaces S) attribute states to sites below the reflection of the initial wave.

Remark. Recall the internal clock phenomenon (cf. end of paragraph 3.1): one unit of time out of two can be, for a particular machine, the moment of the arrival of the reflection of the initial wave. With the above transition rules, this phenomenon is translated by the following fact: *up to the arrival of the reflection of the initial wave, any machine K stays in a particular state an even number of time units.*

This fact, joined to the diagonal propagation of the s_j 's, implies the staircase aspect of the distribution of states on sites below the reflection (cf. Fig. 12).

4.5. Now we consider the reflection of the initial wave and its propagation up to its meeting with a signal S_i under the hypothesis that the created line is not too short (i.e., of length at least 3).

Machine K knows (at time t) that it is (at time $t+1$) upon to deal with the reflection of the initial wave (of the very initial line or of some of the created lines)

if and only if

- (K, t) is in state L;
- $(K + 1, t)$ is in state X or G;
- $(K - 1, t)$ is in state A, B or C.

Moreover, if $(K - 1, t)$ is in state A (respectively, B, C), it does know that the next general to be created will become active—after its creation— with delay 0 (respectively, 1, 2).

All this information will be transmitted by machine K to machine $K - 1$ and will progress along diagonals up to the machine which has to become a general. In order not to introduce new states, we shall code this information as follows:

– the order of appearance of states A, B, C (which is ...ABC..., horizontally and vertically) is broken (hence, inverted since they are only three). This introduces the following transition rules:

$$\begin{aligned} (A, L, X) &\rightarrow C, & (B, L, X) &\rightarrow A, & (C, L, X) &\rightarrow B, \\ (A, L, G) &\rightarrow C, & (B, L, G) &\rightarrow A, & (C, L, G) &\rightarrow B; \end{aligned}$$

– the previous inversion is transmitted to machines along a diagonal using the transition rules:

$$(A, A, C) \rightarrow C, \quad (B, B, A) \rightarrow A, \quad (C, C, B) \rightarrow B.$$

Due to the staircase distribution of states (cf. the Remark in paragraph 4.4), we need not consider environments different from AAC, BBA and CCB. In this way we have set up the reflection of the initial wave. Now we shall deal with the propagation of the information on the delay: it is convenient to introduce an eighth state, denoted R, to mark the end of the delay-transmission process. The three possible delays and the staircase distribution of states lead to the three following sets of rules (cf. Fig. 9). Recall that delay 0 (respectively, 1, 2) is conveyed by s_1 (respectively, s_2, s_3) which is coded by .AB (respectively, .BC, .CA) and at reflection gives .AC (respectively, .BA, .CB).

• *Delay 0:*

$$\begin{aligned} \left. \begin{aligned} (A, C, X) &\rightarrow R, \\ (A, C, G) &\rightarrow R, \end{aligned} \right\} & \text{the right-end machine enters state R after one unit of} \\ & \text{time;} \\ (A, C, R) &\rightarrow R, & \text{propagation of the delay information;} \\ \left. \begin{aligned} (C, R, X) &\rightarrow L, \\ (C, R, G) &\rightarrow L, \\ (C, R, L) &\rightarrow L, \\ (R, L, X) &\rightarrow L, \\ (R, L, G) &\rightarrow L, \\ (R, L, L) &\rightarrow L, \\ (L, L, G) &\rightarrow L, \end{aligned} \right\} & \text{after the passage of the diagonal of R's, any machine} \\ & \text{goes back to the quiescent state L (except in the case} \\ & \text{when the left neighbour is the general); the last rule} \\ & \text{(L, L, G) } \rightarrow L \text{ gives the persistency of L after the passage} \\ & \text{of the R's.} \end{aligned}$$

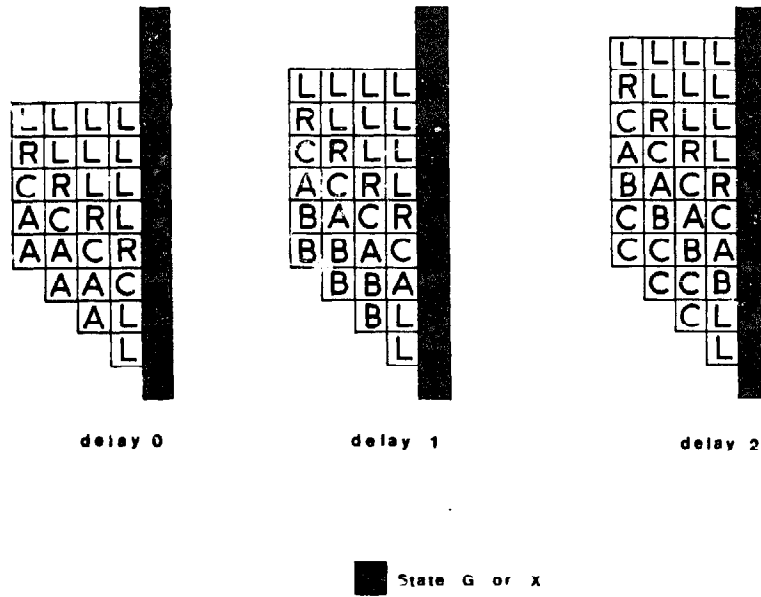


Fig. 9.

• *Delay 1:*

- $$\begin{array}{ll}
 (B, A, X) \rightarrow C, & \text{the right-end machine meets the conditions of delay 0} \\
 (B, A, G) \rightarrow C, & \text{after one unit of time so that the total delay is one;} \\
 (B, A, C) \rightarrow C, & \text{propagation of the delay information;}
 \end{array}$$

plus all the transition rules described in the delay-0 case.

• *Delay 2:*

- $$\begin{array}{ll}
 (C, B, X) \rightarrow A, & \text{the right-end machine meets the conditions of delay 1} \\
 (C, B, G) \rightarrow A, & \text{after one unit of time so that the total delay is two;} \\
 (C, B, A) \rightarrow A & \text{propagation of the delay information;}
 \end{array}$$

plus all the transition rules described in the delay-1 case.

4.6. We now consider the meeting of the reflection of the initial wave with any signal S_i . If, at time t , machine K is in state A (respectively B, C), if its left neighbour is in state L and its right neighbour in state C (respectively A, B) instead of A or B (respectively, B or C, C or A), then machine K knows:

- $(K-1, t)$ is on some S_i ;
- $(K+1, t)$ transmits the reflection of the initial wave.

Hence, it realizes that it has to become a general of a new line at time $t+1$. This introduces the following transition rules:

$$(L, A, C) \rightarrow G, \quad (L, B, A) \rightarrow G, \quad (L, C, B) \rightarrow G.$$

This new general will stay in state G up to the firing time. We shall study in paragraph 4.8 this G -stability question.

The diagonals which transmit the reflection and the delay (there are either 2 or 3 or 4 of such diagonals according to either the value 0 or 1 or 2 of the delay) crash on the vertical line of G's sites. The new general becomes active when its right neighbour is in state R.

A simple analysis of the three types of reflection (associated with the three delays) leads to Fig. 10, and shows that only three rules are necessary:

$(G, A, C) \rightarrow C,$ $(G, C, R) \rightarrow R,$ $(G, R, L) \rightarrow H.$

Observe that after the arrival of the R's diagonal, the new general emits the initial wave of the new right line: in fact, states G and H mark the first and second machines of the new line while other machines of this new line are in the quiescent state.

4.7. We now consider the case where the created line has length two: in this case, the emission of the reflected wave does interfere with its crash on the G's vertical

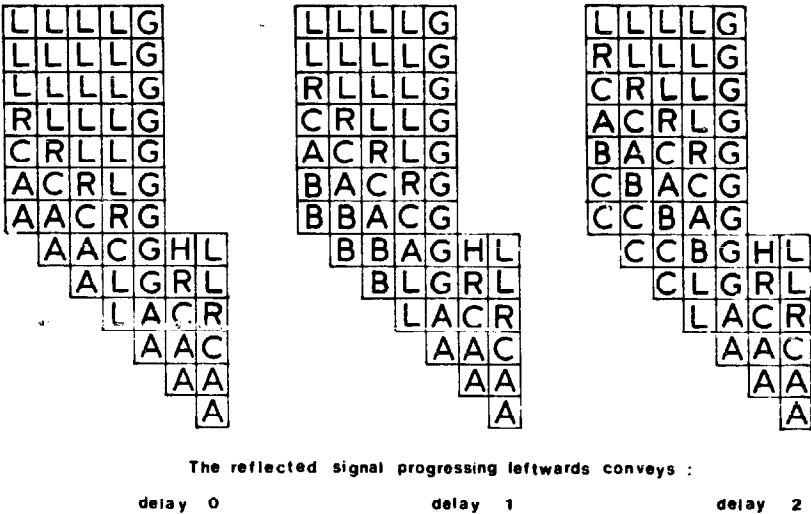


Fig. 10(a). The reflected signal coming from the right conveys delay 0.

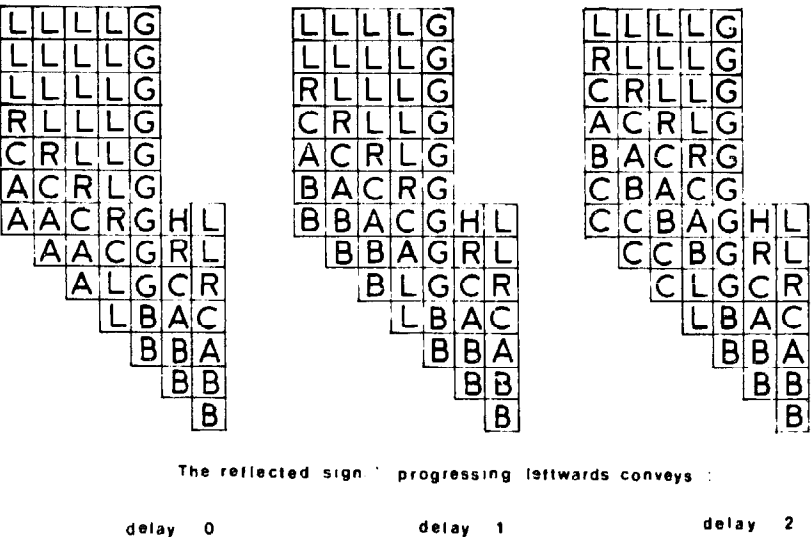


Fig. 10(h). The reflected signal coming from the right conveys delay 1.

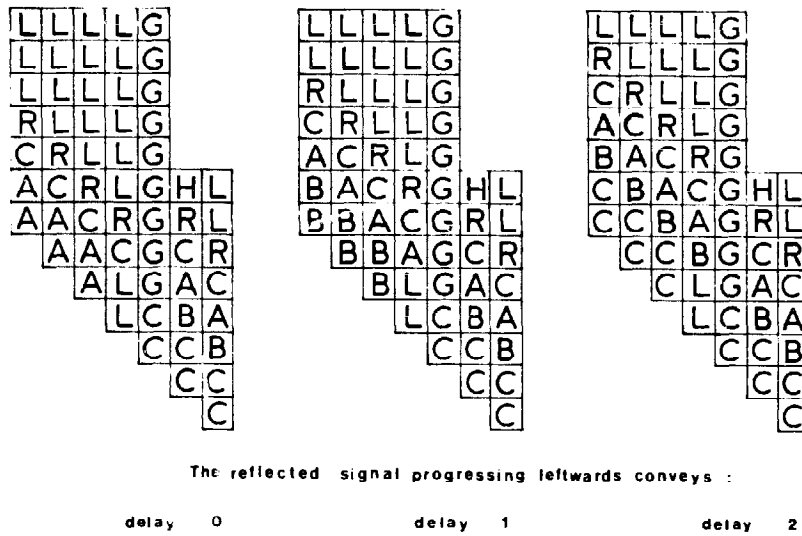


Fig. 10(c). The reflected signal coming from the right conveys delay 2.

line. We introduce four rules:

$(G, A, X) \rightarrow C$, $(G, A, G) \rightarrow C$ these rules complete the previous rules
 $(G, A, C) \rightarrow C$ and $(B, A, G) \rightarrow C$;

$(G, C, X) \rightarrow R$, $(G, C, G) \rightarrow R$ these rules complete the previous rules
 $(G, C, R) \rightarrow R$ and $(A, C, G) \rightarrow R$.

We do not consider environments GB. since they do not appear.

4.8. We first study the stability of G up to the creation of too short lines.

This G-stability question is twofold:

- during the crash of the reflected signal,
- after the created line is operative.

The G-stability in the reflection period leads to nine cases corresponding to:

- the three possible delays conveyed by the reflected signal coming from the right up to the *activation* of the new line,
- the three possible delays conveyed by the reflected signal progressing leftwards along the remaining portion of the initial line (after the *creation* of the new line).

This leads to Fig. 10 which shows the sole environments which have to be considered, and gives the ten following rules:

$(L, G, A) \rightarrow G$, $(L, G, R) \rightarrow G$,
 $(R, G, R) \rightarrow G$ $(A, G, R) \rightarrow G$,
 $(A, G, C) \rightarrow G$, $(B, G, R) \rightarrow G$,
 $(B, G, C) \rightarrow G$, $(C, G, R) \rightarrow G$,
 $(C, G, C) \rightarrow G$, $(L, G, C) \rightarrow G$.

After the new line is activated and up to time $2n-3$, it holds that:

- the right neighbour of the general is always in state L or H,
- the left neighbour of the general can be in any state except state G.

This leads to the following rules:

$$\begin{aligned}
 (A, G, L) &\rightarrow G, & (B, G, L) &\rightarrow G, & (C, G, L) &\rightarrow G, \\
 (R, G, L) &\rightarrow G, & (L, G, L) &\rightarrow G, & (A, G, H) &\rightarrow G, \\
 (B, G, H) &\rightarrow G, & (C, G, H) &\rightarrow G, & (R, G, H) &\rightarrow G, \\
 (L, G, H) &\rightarrow G, & (H, G, L) &\rightarrow G, & (H, G, H) &\rightarrow G.
 \end{aligned}$$

4.9. Finally, we now have to deal with short lines which cannot be broken. A priori, such lines have length 1, 2 or 3.

Since we do not consider an initial line of length 1 and since the breaking process creates lines with length at least 2 (if $n = 3p + i$, where $p \geq 1$ and $i \in \{1, 2, 3\}$, then the new general is machine $2p + i$ so that the line has length $p + 1$ and $p + 1 \geq 2$), we only have to consider short lines of length two or three. Such lines appear in three different contexts:

- the initial line has length two or three,
- a created line has length two or three,
- a remaining portion of a line has length 2 or 3.

The firing synchronization will be obtained as follows:

- up to time $2n-3$, no two adjacent machines are both in state G,
- at time $2n-2$, all machines are in state G,
- at time $2n-1$, all machines are in state F.

The G-synchronization at time $2n-2$ leads to Fig. 11 and to the following transition rules:

$$\begin{aligned}
 (G, L, X) &\rightarrow G, & \text{rule necessary for an initial line of two machines} \\
 & & \text{(Case } a_2); \\
 \left. \begin{aligned} (H, L, X) &\rightarrow H, \\ (G, H, H) &\rightarrow G, \\ (H, H, X) &\rightarrow G, \end{aligned} \right\} & \text{rules necessary for an initial line of three machines} \\
 & & \text{(Case } a_3); \\
 (G, R, X) &\rightarrow G, & \text{complementary rule necessary for rightmost created} \\
 & & \text{lines of length 2 (Case } b_2).
 \end{aligned}$$

No new rule is needed for the case of a rightmost created line of length 3 (Case b_3).

$$\begin{aligned}
 (G, R, G) &\rightarrow G, & \text{complementary rule necessary for the generic-created} \\
 & & \text{line of length 2 (Case } c_2); \\
 \left. \begin{aligned} (H, L, G) &\rightarrow H \\ (H, H, G) &\rightarrow G \end{aligned} \right\} & \text{complementary rules necessary for the generic created} \\
 & & \text{line of length 3 (Case } c_3); \\
 (G, L, G) &\rightarrow G, & \text{complementary rule necessary for the remaining} \\
 & & \text{portion of a line, with length 2 (Case } d_2).
 \end{aligned}$$

No new rule is needed for the case illustrated by Case d_3 .

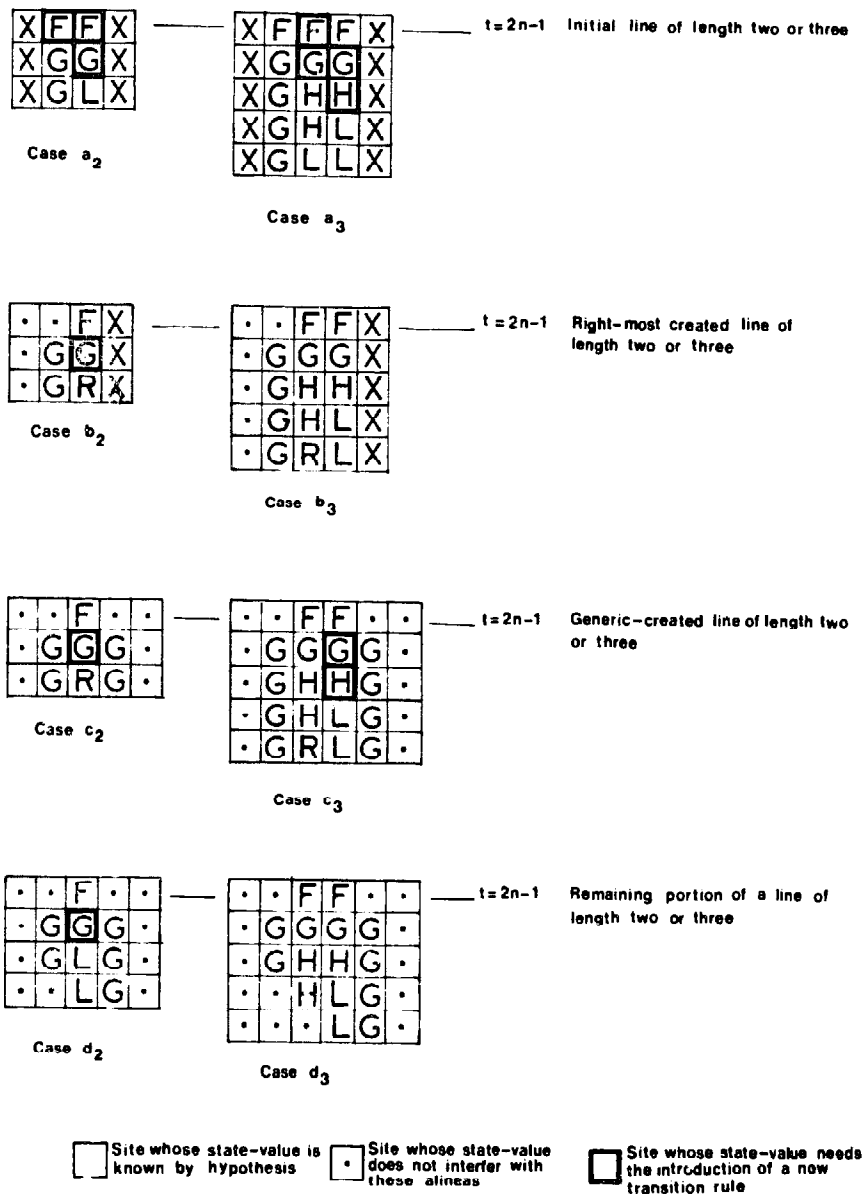


Fig. 11.

Lastly we consider the stability of G within the context of short lines. Fig. 11 shows that at time $2n-3$, all machines are in state G, H, R, L. To complete the stability of G (from time $2n-3$ to time $2n-2$) it suffices to introduce a single new rule: $(H, G, R) \rightarrow G$.

With these rules all short lines are G-synchronized at time $2n-2$. The whole initial line, being covered by these short lines, is thus G-synchronized at time $2n-2$.

It is now clear that the firing synchronization is ensured by the three following rules:

$$(G, G, G) \rightarrow F, \quad (X, G, G) \rightarrow F, \quad (G, G, X) \rightarrow F.$$

4.10. Now we have all necessary transition rules which fill the seven matrices of

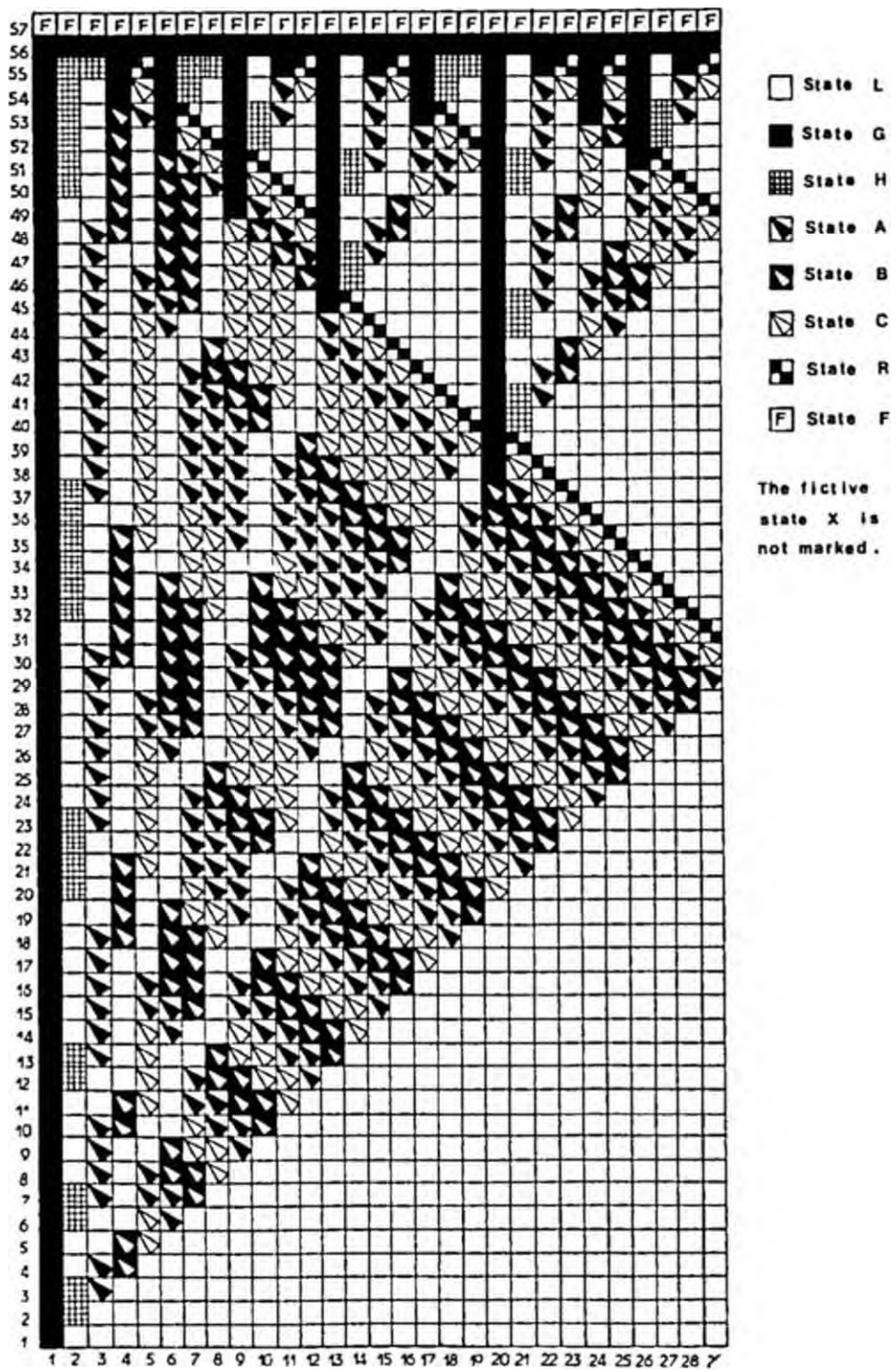


Fig. 12. Observe that (i): machine 20 becomes a general at time 38 and active at time 39 (delay 1); (ii): machine 13 becomes a general at time 45 and active at time 45 (delay 0); (iii): machine 9 becomes a general at time 49 and active at time 51 (delay 2).

Fig. 13. Observe that no two rules are contradictory. Also observe that the blanks in the matrices correspond to environments which are not met, hence for which no transition rule is necessary.

Figure 12 shows the synchronization of an initial line of 29 machines.

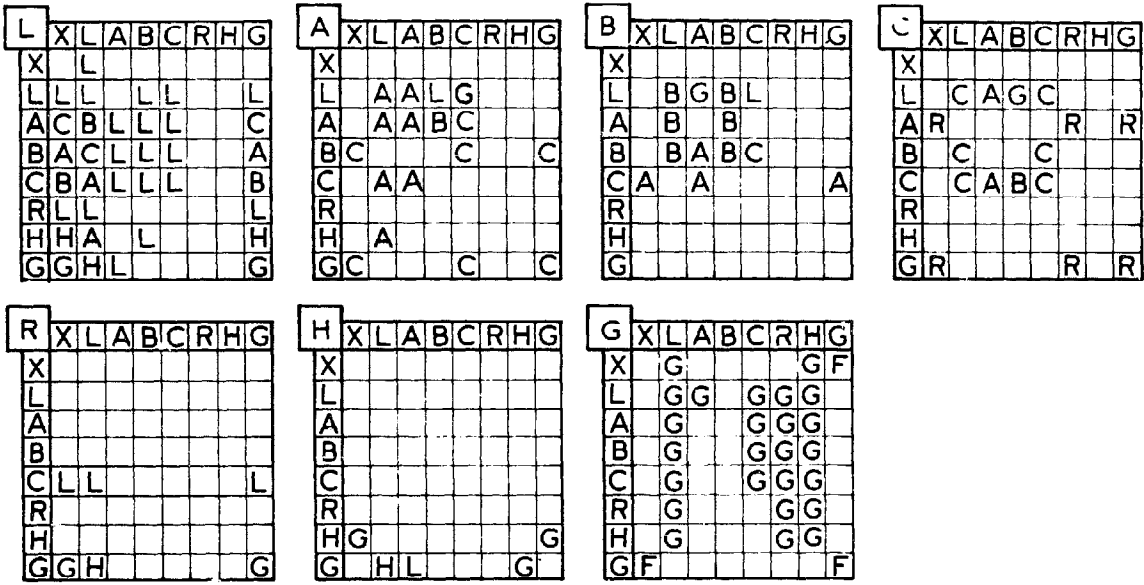


Fig. 13.

5. A six-state minimal time solution

5.1. In this section we shall show how to eliminate two states in the preceding solution. It is easy to eliminate state H. The matrices of the 8-state solution show that state R appears very few times. We are first going to eliminate state R and then state H.

To do this, we shall introduce new rules, some of them, marked by (*), contradicting old ones. To get over the contradiction, we shall also abandon some old rules and introduce some new ones. And so on, up to an equilibrium.

5.2. Observe that no B-diagonal belongs to the part of the reflection which crashes on a G-vertical line (cf. Fig. 12). This leads us to replace state R by state B and leads to the following transitions rules (suggested by Figs. 7-13):

$$\begin{aligned}
 & (B, L, L) \rightarrow L, \quad (*) \quad (B, L, X) \rightarrow L, \quad (*) \quad (B, L, G) \rightarrow L, \quad (*) \\
 & (C, B, X) \rightarrow L, \quad (*) \quad (C, B, G) \rightarrow L, \quad (*) \\
 & (A, C, X) \rightarrow B, \quad (A, C, G) \rightarrow B, \quad (C, B, L) \rightarrow L, \\
 & (G, B, L) \rightarrow H, \quad (G, B, G) \rightarrow G, \quad (G, B, X) \rightarrow G, \\
 & (A, C, B) \rightarrow B, \quad (G, C, B) \rightarrow B, \quad (G, C, G) \rightarrow B. \\
 & (G, C, X) \rightarrow B, \quad (H, G, B) \rightarrow G,
 \end{aligned}$$

Unfortunately, we have that

- rules marked by (*) contradict rules

$$(B, L, L) \rightarrow C, \quad (B, L, X) \rightarrow A, \quad (B, L, G) \rightarrow A;$$

- rules marked by (*2) contradict rules

$$(C, B, X) \rightarrow A, \quad (C, B, G) \rightarrow A.$$

5.3. In order to maintain rules marked by (*1) we suppress the environment BLL below the reflection of the initial wave. To do this we modify the attribution of state B to sites on a signal s_2 (between S_i and S_{i+1}) and to sites between s_2 and the preceding signal s_1 . Using the internal clock phenomenon we replace the portion of the state-diagram BBL by BCL and BLI by GLL. In this way, we also suppress the environments BLX and BLG occurring on the reflection of an initial wave. Looking at Figs. 12 and 7, we see that this leads to

- the suppression of

$$\begin{aligned} (B, L, L) \rightarrow C, \quad (A, L, L) \rightarrow B, \quad (B, B, L) \rightarrow B, \\ (B, L, G) \rightarrow A, \quad (B, L, X) \rightarrow A, \quad (A, B, L) \rightarrow B, \\ (L, B, L) \rightarrow B. \end{aligned}$$

- the introduction of

$$\begin{aligned} (L, G, L) \rightarrow B, \quad (*3) \quad (G, L, L) \rightarrow C, \quad (*4) \\ (A, L, L) \rightarrow G, \quad (B, B, L) \rightarrow G, \quad (A, A, G) \rightarrow B, \\ (L, B, L) \rightarrow G, \quad (B, B, G) \rightarrow B, \quad (A, G, L) \rightarrow B, \quad (*3) \\ (G, L, B) \rightarrow L, \quad (G, L, C) \rightarrow L, \quad (L, B, G) \rightarrow B, \\ (B, G, L) \rightarrow B, \quad (*3) \quad (G, L, G) \rightarrow A, \quad (*3) \quad (G, L, X) \rightarrow A. \quad (*4) \end{aligned}$$

Unfortunately, rules marked by (*4) and (*3) contradict the following rules:

$$\begin{aligned} (L, G, L) \rightarrow G, \quad (G, L, L) \rightarrow H, \quad (G, L, G) \rightarrow G, \\ (A, G, L) \rightarrow G, \quad (B, G, L) \rightarrow G, \quad (G, L, X) \rightarrow G. \end{aligned}$$

5.4. In order to maintain rules marked by (*2), we modify the reflection of the initial wave in the case of delay 2. To do this, we replace the portions of diagrams

$$\begin{array}{ccc} B A C Z & & B A C Z \\ C B A Z & & C B A Z \\ C C B Z & \text{by} & C C G Z \\ C L Z & & C L Z \end{array}$$

where Z is state X or G . In this way we suppress the environments CBX and CBG.

Looking at these parts of the diagrams and Fig. 9 (for the reflected wave), we see that this leads to

- the suppression of

$$\begin{aligned} (C, L, X) \rightarrow B, \quad (C, B, X) \rightarrow A, \\ (C, L, G) \rightarrow B, \quad (C, B, G) \rightarrow A; \end{aligned}$$

- the introduction of

$$\begin{aligned} (C, L, X) \rightarrow G, & \quad (C, G, X) \rightarrow A, & \quad (C, C, G) \rightarrow B, \\ (C, L, G) \rightarrow G, & \quad (C, G, G) \rightarrow A, & \quad (G, H, G) \rightarrow G. \\ (G, G, C) \rightarrow G, & \quad (G, G, B) \rightarrow G, \end{aligned}$$

In the case that the line to be created is going to be short, the preceding diagrams have to be modified. To suppress the environments CBX and CBG we have to replace the portions of diagrams

$$\begin{array}{ccc} LGAZ & & LGAZ \\ LCBZ & \text{by} & LCGZ \\ CLZ & & CLZ \end{array}$$

where Z is state X or G . This introduces the rule $(L, C, G) \rightarrow G$.

5.5. Observe that the environments of the rules marked by (*3) occur only when machine 2 is in state L . In order to maintain these rules marked by (*3), we always (in fact, up to time $2n - 2$) place machine 2 in state H . To do this, we replace the two portions of diagrams:

$$\begin{array}{ccc} & GHLB & \\ GLA & GHLB & \\ GLA & GLLB & \\ GHA & GLAB & \end{array} \quad \text{by} \quad \begin{array}{ccc} & GHLB & \\ GHA & GHLB & \\ GHA & GHLB & \\ GHA & GHAB & \end{array}$$

This leads to

- the suppression of rules

$$\begin{aligned} (H, G, L) \rightarrow G, & \quad (G, H, A) \rightarrow L, \\ (L, G, L) \rightarrow G, & \quad (A, G, L) \rightarrow G, & \quad (B, G, L) \rightarrow G, \\ (C, G, L) \rightarrow G; \end{aligned}$$

- the introduction of rules

$$(G, H, A) \rightarrow H, \quad (H, A, B) \rightarrow L.$$

This new state-value introduced for machine 2 also modifies the G -synchronization in Case d_2 (see Fig. 11). This leads to the suppression of rule $(G, L, G) \rightarrow G$ and to the introduction of rule $(G, H, G) \rightarrow G$.

Now there only remain two contradictory rules: $(G, L, X) \rightarrow A$ and $(G, L, L) \rightarrow C$. We observe that the rule $(G, L, X) \rightarrow G$ is used only when the initial line is of length 2: when the rightmost new line is of length 2, the rule $(G, B, X) \rightarrow G$ introduced in paragraph 5.2 is used.

We observe that rule $(G, L, L) \rightarrow H$ is used only to set up machine 2. Thus, the two remaining contradictions concern only site $(2, 2)$.

5.6. Now we eliminate state H; this suppresses rule $(G, L, L) \rightarrow H$. Rule $(G, B, G) \rightarrow G$ (respectively $(G, L, L) \rightarrow A$) suggests us to replace state H by state B (respectively C). In fact, using the internal clock phenomenon we shall replace state H alternatively by states C and B. Observe (look at site (3, 21) in Fig. 14) that we must introduce the rule $(C, L, G) \rightarrow L$ which contradicts the rule $(C, L, G) \rightarrow G$ introduced in paragraph 5.4. To avoid this contradiction, we shall code states of machine 3 in a particular way: machine 3 will be alternatively in state A and in state G. To do this, we replace the diagrams of Fig. 14(a) by the diagrams occurring in Fig. 14(b).

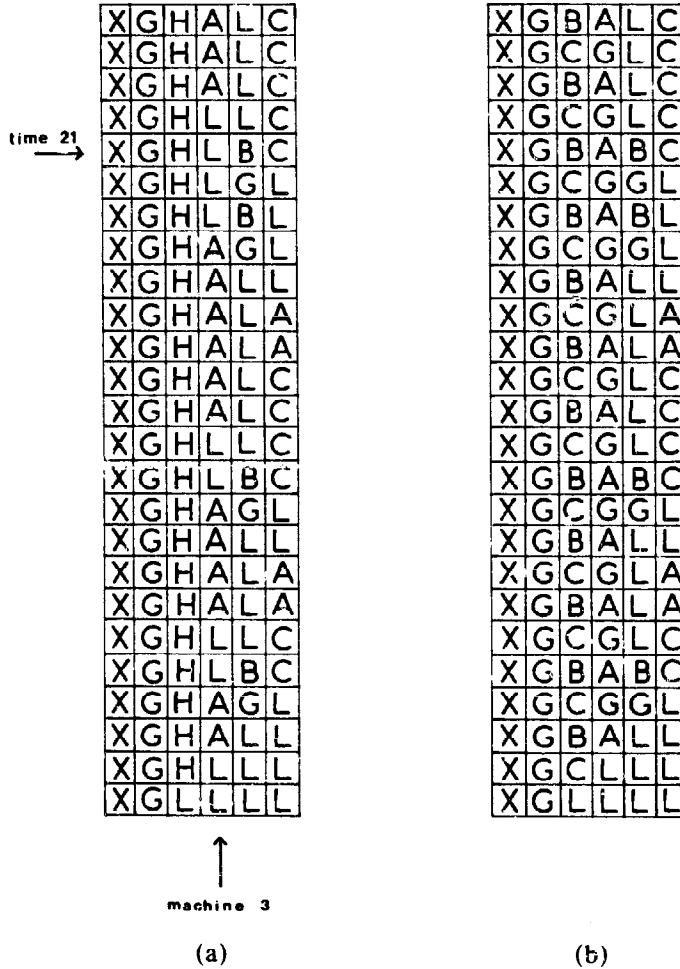


Fig. 14.

This leads to

- the suppression of all rules in which a state H occurs and of the following rules:

$$(L, B, L) \rightarrow G, \quad (L, G, L) \rightarrow B, \quad (L, A, L) \rightarrow A;$$

- the introduction of the following rules:

$$(G, L, L) \rightarrow C, \quad (G, C, L) \rightarrow B, \quad (G, B, A) \rightarrow C,$$

$$(G, C, G) \rightarrow B, \quad (B, A, L) \rightarrow G, \quad (G, B, L) \rightarrow C,$$

$$\begin{aligned}
& (B, A, B) \rightarrow G, \quad (C, G, L) \rightarrow A, \quad (G, G, L) \rightarrow B, \\
& (A, B, C) \rightarrow L, \\
& (A, B, L) \rightarrow G, \quad (X, G, B) \rightarrow G, \quad (X, G, C) \rightarrow G.
\end{aligned}$$

These new site values for 2 and 3 modify the G-synchronization, when:

- the initial line is of length 3 or 4 (Cases a_3 , a_4 of Fig. 11);
- the remaining portion of the initial line is of length 3 or 4 (Cases c_3 , c_4 of Fig. 11).

This leads to the introduction of the following rules:

$$\begin{aligned}
& (A, B, A) \rightarrow B, \quad (G, B, C) \rightarrow B, \quad (B, C, G) \rightarrow G, \\
& (B, G, G) \rightarrow G, \quad (B, G, X) \rightarrow G, \quad (B, C, X) \rightarrow G.
\end{aligned}$$

In this way we obtain a six-state automaton which synchronizes all lines consisting of more than two machines.

The special case of an initial line of two machines is solved by the two following diagrams:

$$\begin{array}{cc}
XFFX & XGBL \\
XAA X & XACL \\
XGLX & XGLL
\end{array}$$

The first diagram ensures the synchronization of an initial line of length 2; the second one deals with the consequences brought by the first. This leads to the introduction of the following rules:

$$\begin{aligned}
& (A, C, L) \rightarrow B, \quad (X, A, C) \rightarrow G, \\
& (X, G, L) \rightarrow A, \quad (X, A, A) \rightarrow F, \quad (A, A, X) \rightarrow F
\end{aligned}$$

and the suppression of the rule $(X, G, L) \rightarrow A$.

5.7. Observe that these modified rules have the following consequence: a site (K, t) in S_i receives state L if and only if $K \geq 4$ (contrary to the situation described in Section 4; this is due to the particular coding device of machines 2 and 3).

5.8. Now we have all necessary transition rules which fill the five matrices of Fig. 15. Observe that no two rules are contradictory. Also observe that the blanks in the matrices correspond to environments which are not met; hence, for which no transition rule is necessary. Figure 16 shows the synchronization of an initial line of 29 machines.

6. Proof of correctness

6.1. We now prove that the automaton defined in Section 5 (whose transition rules are indicated in Fig. 15) is indeed a minimal time solution of the firing squad synchronization problem.

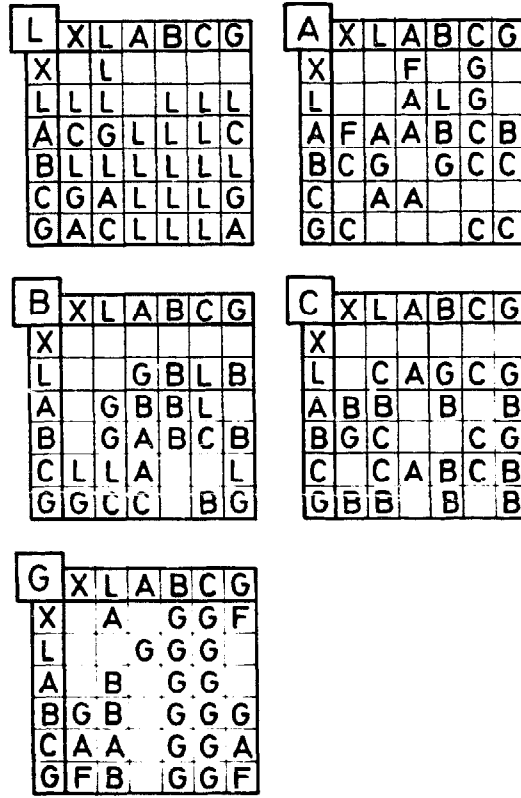


Fig. 15.

The proof will proceed by induction on the length n of the initial line: the synchronization of a line of length $n = 3p + i$ (with $i \in \{1, 2, 3\}$) is reduced to the synchronization of lines of length $2p + i - 1$ and $p + 1$ (corresponding to the break at $\frac{2}{3}$ described in Section 2). In order to get such a reduction, we first prove some facts relative to

- the behaviour of the line below the reflection (the scheme),
- the reflection itself.

The following fact will be useful:

6.1. Fact Every site (K, t) with $K > t$ has state L.

Proof. This is a trivial consequence of the rules $(L, L, L) \rightarrow L$ and $(L, L, X) \rightarrow L$. $\square$

Finally, we shall often use the fact that the distribution of states on a set X fully determines that on a set $\tilde{X}$ bigger than X as illustrated by Fig. 17.

6.2. It is clear (cf. the study of signals s_j in Section 3) that diagonals are the essential objects to be considered. Due to the internal clock phenomenon, the most pertinent object is that of *double diagonals* DD_m :

6.2. Notation. If $7 \leq m < n$, we let DD_m be the set

$$\{(m-i, m+i), (m-i, m+i+1) \mid i \in \{-1, 0, \dots, m-1\}\}$$

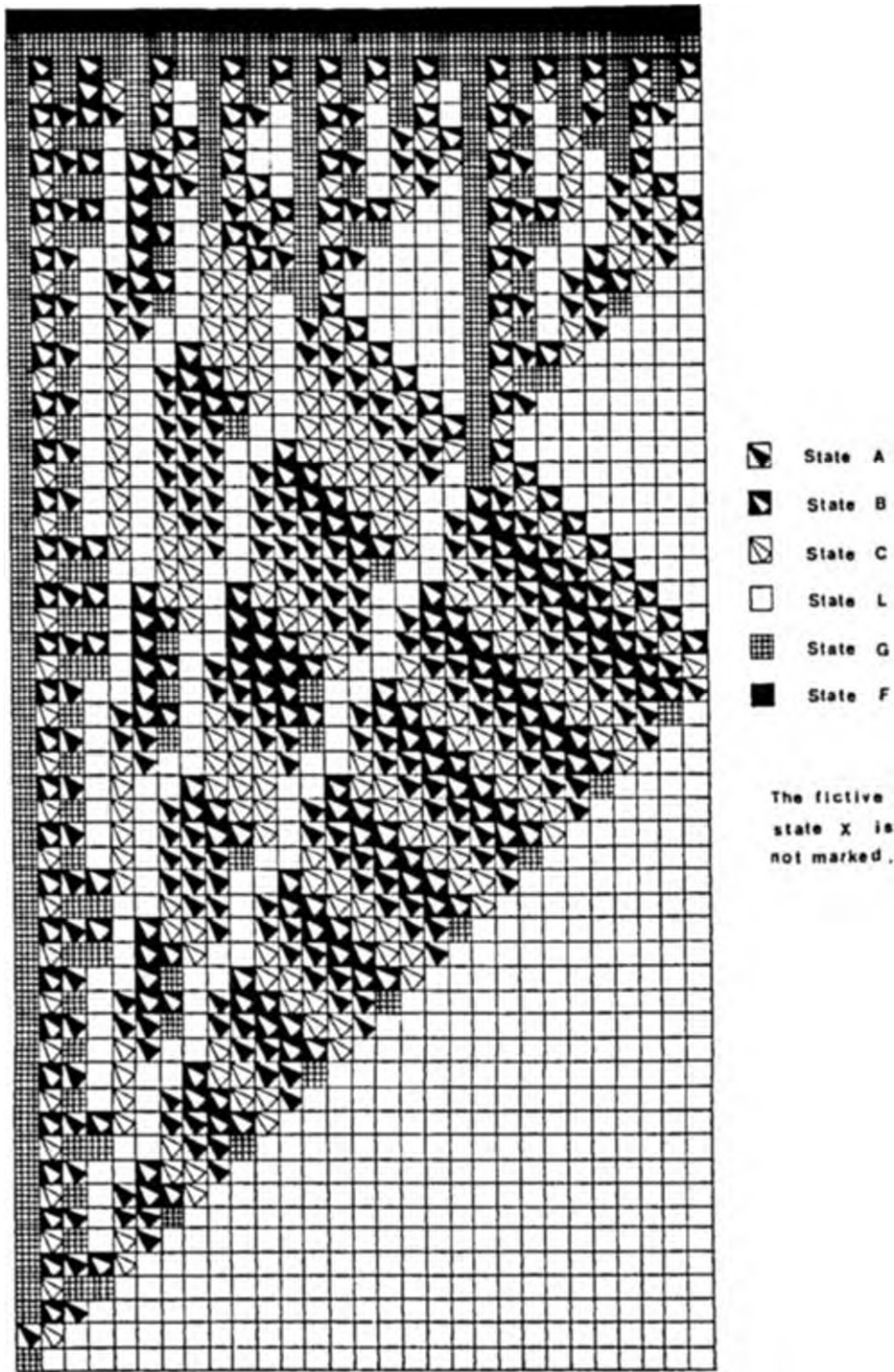


Fig. 16. Observe that (i): machine 20 becomes a general at time 38 and active at time 39 (delay 1); (ii): machine 13 becomes a general at time 45 and active at time 45 (delay 0); (iii): machine 9 becomes a general at time 49 and active at time 51 (delay 2).

(i is allowed to be -1 in order that DD_m captures two sites corresponding to machine $m+1$).

As can be seen in Fig. 16, a double diagonal DD_m consists of several 'unicolour' pieces. This leads us to introduce *truncated* double diagonals.

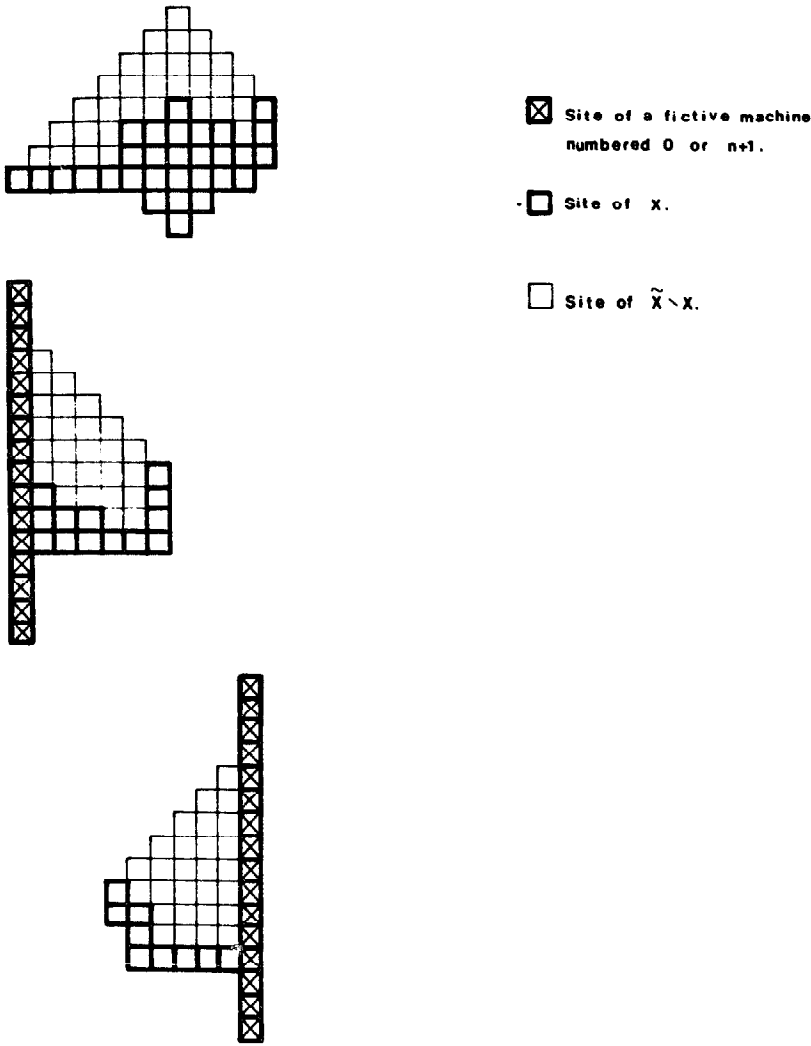


Fig. 17.

6.3. Notation. If $7 \leq m < n$ and $1 \leq K < j \leq m+1$, we let $DD_{m,j,K}$ be the set

$$\begin{aligned}
 & DD_m \cap \{(l, t) \mid K \leq l \leq j\} \\
 & = \{(m-i, m+i), (m-i, m+i+1) \mid i \in \{m-j, \dots, m-K\}\}.
 \end{aligned}$$

Remark. Observe that indices m, j, K are in decreasing order: this is indeed the order of appearance of machines when one progresses upward on DD_m .

We distinguish some families of basic truncated double diagonals.

6.4. Definition. Suppose $j \geq K+3$ and $K \geq 2$.

- (1) We say that $DD_{m,j,K}$ is A-basic (respectively C-basic) if
- (α) its rightmost and leftmost sites $(j, 2m-j)$, $(j, 2m-j+1)$ and $(K, 2m-K)$, $(K, 2m-K+1)$ have state L;
 - (β) its other sites $(i, 2m-i)$, $(i, 2m-i+1)$ have state A (respectively C) (with i in $\{m-j+1, \dots, m-K-1\}$).

- (2) We say that $DD_{m,j,K}$ is B-basic if condition (α) above is satisfied and
 (β') its left to rightmost sites $(j-1, 2m-j+1)$, $(j-1, 2m-j)$ have states B
 and G: its other sites $(i, 2m-i)$, $(i, 2m-i+1)$ (with i in $\{m-j+2, \dots, m-K+1\}$) have state B.
- (3) $DD_{m,j,K}$ is basic if it is A-, B-, or C-basic.

Remark. The reason for the condition $j \geq K+3$ is that the extreme sites are distant enough and do not directly interfere.

The next two lemmas describe some possible evolutions in time of truncated double diagonals (up to the arrival of the reflected signal).

6.5. Lemma. *Let $j \geq K+3$ and $K \geq 2$.*

- (1) *If $DD_{m,j,K}$ is B-basic (respectively C-basic) and if the rightmost sites of $DD_{m+1,j,K}$ (i.e., $(j, 2m-j+2)$ and $(j, 2m-j+3)$) have state L, then $DD_{m+1,j,K}$ is also B-basic (respectively C-basic).*
- (2) *If $DD_{m,j,K}$ is A-basic and the rightmost sites of $DD_{m+1,j,K}$ have state L and if
 sites $(K-1, 2m-K+1)$ and $(K-1, 2m-K+2)$
 have states different from L,* (5)

then $DD_{m+1,j,K}$ is also A-basic.

Proof. We refer to Fig. 18(a).

Case C-basic: Using rules $(C, C, L) \rightarrow C$, $(C, C, C) \rightarrow C$, and $(L, C, C) \rightarrow C$ we see that all sites $(i, 2m-i+2)$ (and then all sites $(i, 2m-i+3)$) for $i \in \{K+1, \dots, j-1\}$ are in state C.

Observe that the environment (L, C, L) does not occur: there are at least two sites between the leftmost and rightmost sites of $DD_{m,j,K}$ since $j \geq K+3$. States marked by U , V on Fig. 18(a) (on sites $(K-1, 2m-K+1)$ and $(K-1, 2m-K+2)$) are not X since $K \neq 1$. This fact and the transition rules $(\cdot, L, C) \rightarrow L$ where $\cdot$ is any state different from X, show that $(K, 2m-K+3)$, $(K, 2m-K+2)$ have state L. This proves that $DD_{m+1,j,K}$ is C-basic.

Case B-basic: We proceed in a similar way. Observe that $(B, B, L) \rightarrow G$ attributes state G to site $(j-1, 2m-j+3)$. This imposes us to consider the case $j = K+3$ (see Fig. 18(a)) for which we use rule $(L, B, G) \rightarrow B$.

Case A-basic: We proceed as in the case C-basic. The lack of any transition rule with environment (L, L, A) is supplied by hypothesis (5). $\square$

6.6. Lemma. *Let $j \geq K+3$ and $K \geq 2$ and $l \leq m < n$.*

- (1) *If $DD_{m,j,K}$ is A-basic (respectively B-basic) and if the rightmost sites of $DD_{m+1,j+1,K+1}$ (i.e., sites $(j+1, 2m-j+1)$ and $(j+1, 2m-j+2)$) have state L, then $DD_{m+1,j+1,K+1}$ is B-basic (respectively C-basic).*
- (2) *If $DD_{m,j,K}$ is C-basic, if the rightmost sites $DD_{m+1,j+1,K}$ have state L and if condition (5) of Lemma 6.5 is satisfied, then $DD_{m+1,j+1,K}$ is A-basic.*

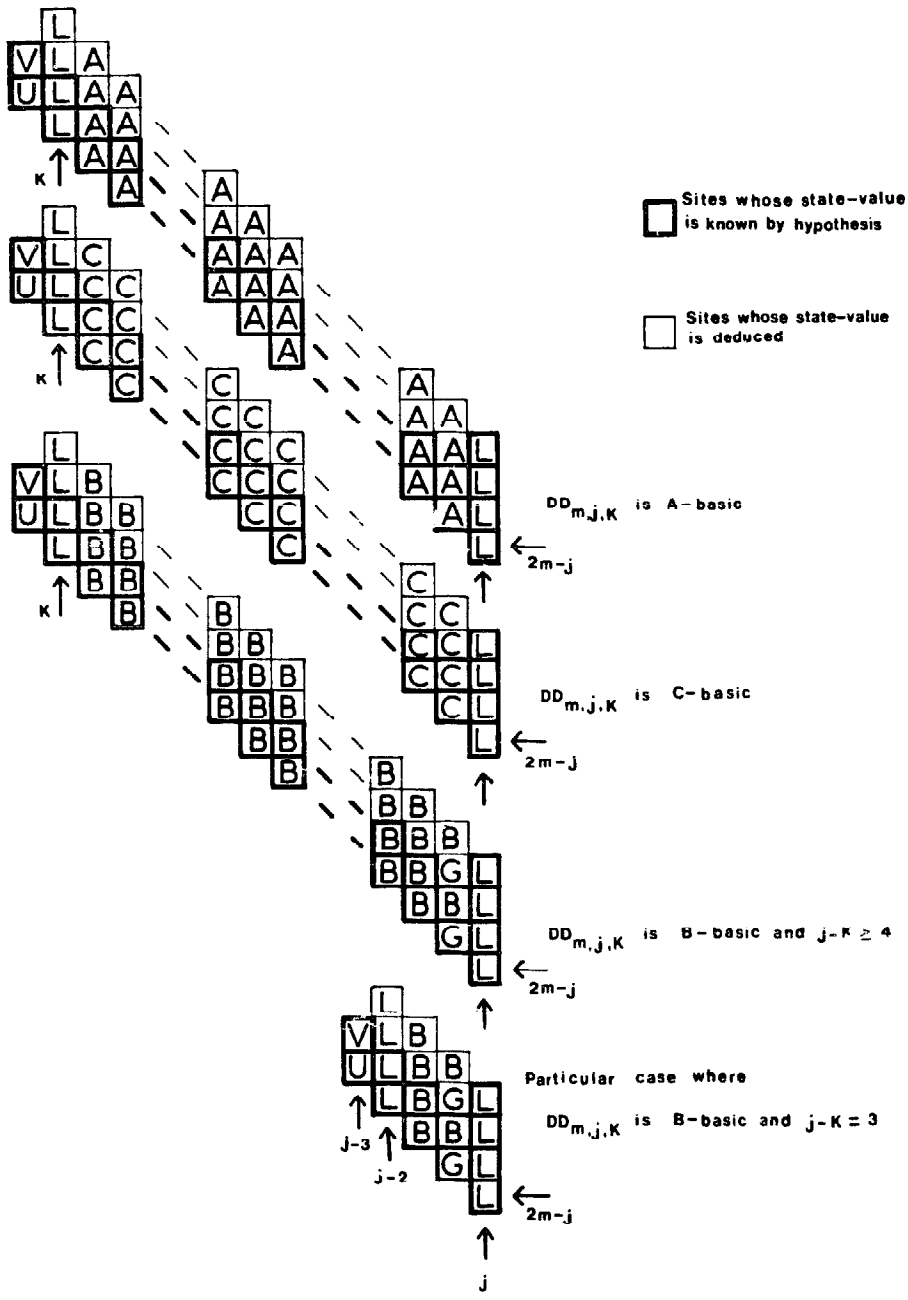


Fig. 18(a).

The proof is similar to that of Lemma 6.5 and is illustrated by Fig. 18(b).

Looking at Fig. 16, we see that double diagonals consist of basic truncated double diagonals and a terminal portion somewhat different.

6.7. Definition. Let $7 \leq m < n$; a truncated double diagonal $DD_{m,4,1}$ (respectively, $DD_{m,5,1}$, $DD_{m,6,1}$) is called a 4-end (respectively 5-end, 6-end) if its site-values are those indicated by Fig. 18(c) Case 1 (respectively, Case 2, Case 3).

6.8. Lemma. Let $7 \leq m < n$. There exists a finite sequence of integers $(j_1^m, \dots, j_{l_m}^m)$

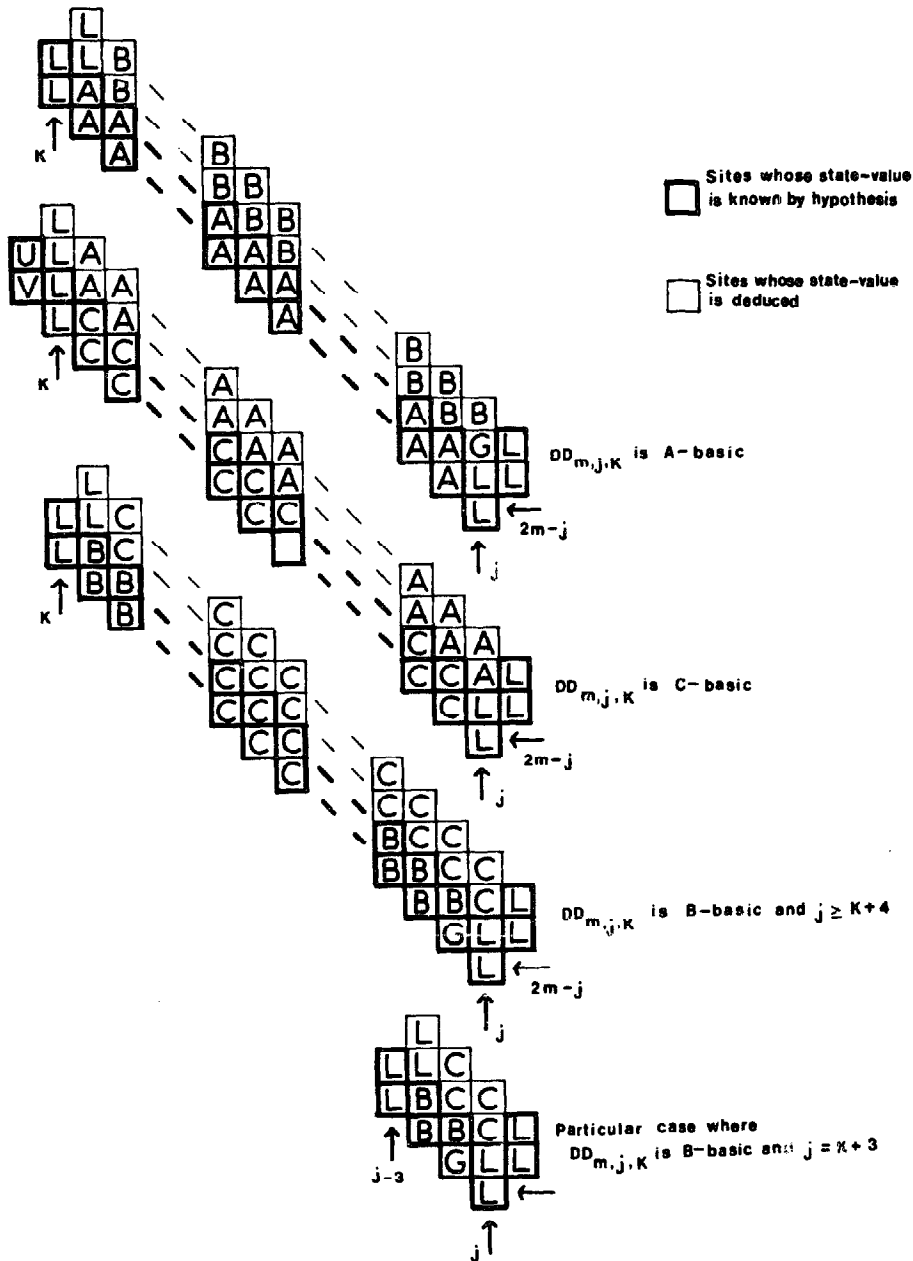


Fig. 18(b).

such that

$$4 \leq j_{l_m}^m \leq \dots \leq j_{l+1}^m \leq j_l^m \leq \dots \leq j_1^m = m+1$$

and

- (1) for every l in $\{1, \dots, l_m - 1\}$, the portion of the double diagonal DD_{m, j_1^m, j_{l+1}^m} is basic.
- (2) The integer $j_{l_m}^m$ is 4, 5, or 6 and the terminal portion $DD_{m, j_{l_m}^m, 1}$ is a $j_{l_m}^m$ -end.

Proof. The argument is by induction on the integer m in the segment $[7, \dots, n-1]$.

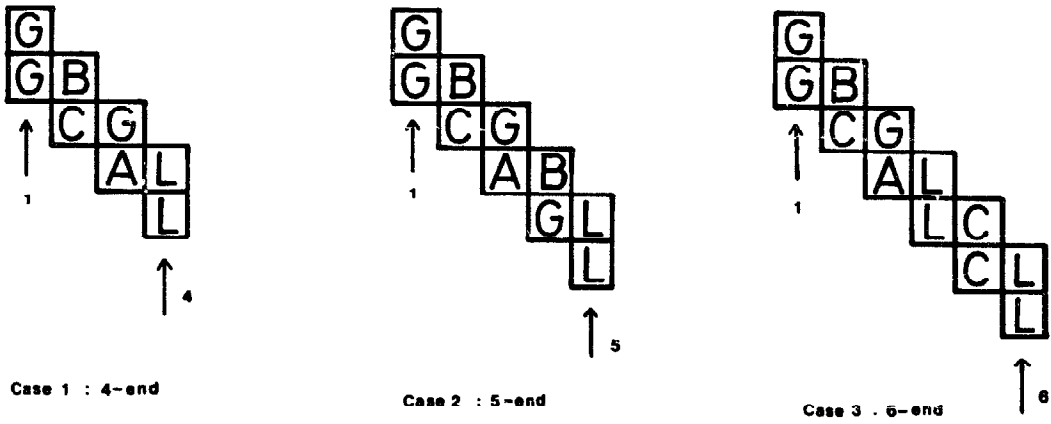


Fig. 18(c).

(i) *Case $m = 7$* : Observing that the site-values of DD_7 are independent of n ($n \geq 8$), it suffices to check this case with any particular value of n . This can be done using Fig. 16.

We now assume that the lemma is true up to m ($m \leq n - 2$) and prove the case $m + 1$. The proof is quite long and occupies (ii) to (v) below.

(ii) We define l_{m+1} and the sequence $j_{l_{m+1}}^{m+1}, \dots, j_l^{m+1}, \dots, j_1^{m+1}$ as follows (I):

- l_{m+1} is $l_m + 1$ if $j_{l_m}^m = 6$ and all truncated double diagonals DD_{m, j_l^m, j_{l+1}^m} (with l in $\{1, \dots, l_m - 1\}$) are A- or B-basic; l_{m+1} is l_m otherwise.
- For l in $\{1, \dots, l_m\}$, j_l^{m+1} is $j_l^m + 1$ if all truncated double diagonals DD_{m, j_s^m, j_{s+1}^m} with $s \in \{1, \dots, l - 1\}$ are A- or B-basic; j_l^{m+1} is j_l^m otherwise; in particular for $l = 1$, $j_1^{m+1} = j_1^m + 1 = m + 2$.
- If $l_{m+1} = l_m + 1$, then $j_{l_{m+1}}^{m+1} = 4$.

We observe that if $j_l^{m+1} = j_l^m$, then $j_s^{m+1} = j_s^m$ for all s in $\{l, \dots, l_m\}$ and $l_{m+1} = l_m$.

(iii) We first observe that condition (5) in Lemmas 6.5 and 6.6 is always satisfied for the DD_{m, j_l^m, j_{l+1}^m} with $l \in \{1, \dots, l_m - 1\}$:

- *Case $l = l_m - 1$* : It is clear from property (2) (in Lemma 6.8) for $DD_{m, j_{l_m}^m, 1}$ and Fig. 18(c) that sites $(j_{l_m}^m - 1, 2m - j_{l_m}^m + 1)$ and $(j_{l_m}^m - 1, 2m - j_{l_m}^m + 2)$ are not in state L.
- *Case $l < l_m - 1$* : It is clear from property (1) (Lemma 6.8) for $DD_{m, j_{l+1}^m, j_{l+2}^m}$ and Definition 6.4 that sites $(j_{l+1}^m - 1, 2m - j_{l+1}^m + 1)$ and $(j_{l+1}^m - 1, 2m - j_{l+1}^m + 2)$ are not in state L (they are both in state A or both in state C or in respective states G, B).

(iv) We first prove a part of property (1) for DD_{m+1} : double diagonals $DD_{m+1, j_l^{m+1}, j_{l+1}^{m+1}}$ are basic for $l \in \{1, \dots, l_m - 1\}$ (the only possibly remaining $DD_{m+1, j_l^{m+1}, j_{l+1}^{m+1}}$ —when $l_{m+1} = l_m + 1$ —will be studied in paragraph (v)). The proof is by induction on l in $\{1, \dots, l_m - 1\}$.

* *Initial step $l = 1$* : Using Fact 6.1, we know that sites $(m + 2, m + 1)$, $(m + 2, m)$ are in state L. Since $m + 1 = j_1^m$, this shows that $DD_{m, m+1, j_2^m}$ satisfies the hypothesis of Lemma 6.6. Noticing that the different cases in Lemma 6.6 correspond to the different cases in the definition of j_2^{m+1} from j_2^m , the conclusion of Lemma 6.6 establishes that $DD_{m+1, m+2, j_2^{m+1}}$ is basic.

* *Induction step*: We suppose that $DD_{m+1, j_l^{m+1}, j_{l+1}^{m+1}}$ is basic (with $1 < l < l_m - 1$). First we observe that the rightmost sites of $DD_{m+1, j_{l+1}^{m+1}, j_{l+2}^{m+1}}$ are the leftmost sites of

$DD_{m+1, j_l^{m+1}, j_l^{m+1}}$. The induction hypothesis (over l) insures that these sites are in state L.

- *Case $j_{l+1}^{m+1} = j_{l+1}^m + 1$:* The very definitions of j_{l+1}^{m+1} from j_{l+1}^m show that every DD_{m, j_s^m, j_{s+1}^m} , with $s \in \{1, \dots, l-1\}$ is A- or B-basic. Thus,

$$j_{l+2}^{m+1} = \begin{cases} j_{l+2}^m + 1 & \text{if } DD_{m, j_{l+1}^m, j_{l+2}^m} \text{ is A- or B-basic,} \\ j_{l+2}^m & \text{if it is C-basic.} \end{cases}$$

Since sites $(j_{l+1}^m + 1, 2m - j_{l+1}^m + 1)$ and $(j_{l+1}^m + 1, 2m - j_{l+1}^m + 2)$ —being the rightmost sites of $DD_{m+1, j_{l+1}^{m+1}, j_{l+2}^{m+1}}$ since $j_{l+1}^{m+1} = j_{l+1}^m + 1$ —have state L, the hypothesis of Lemma 6.6 is satisfied for $DD_{m, j_{l+1}^m, j_{l+2}^m}$.

Observing that the different cases in Lemma 6.6 correspond to the different cases of the definitions of j_{l+2}^{m+1} from j_{l+2}^m , we get the basic character of $DD_{m+1, j_{l+1}^{m+1}, j_{l+2}^{m+1}}$.

- *Case $j_{l+1}^{m+1} = j_{l+1}^m$:* Recall (cf. end of paragraph (ii)) that this equality implies $j_{l+2}^{m+1} = j_{l+2}^m$. Since sites $(j_{l+1}^m, 2m - j_{l+1}^m + 2)$ and $(j_{l+1}^m, 2m - j_{l+1}^m + 3)$ —being the rightmost sites $DD_{m+1, j_{l+1}^{m+1}, j_{l+2}^{m+1}}$ since $j_{l+1}^{m+1} = j_{l+1}^m$ —have state L, the hypothesis of Lemma 6.5 is satisfied for $DD_{m+1, j_{l+1}^{m+1}, j_{l+2}^{m+1}}$.

Thus, $DD_{m+1, j_{l+1}^{m+1}, j_{l+2}^{m+1}}$ —which is the same as $DD_{m+1, j_{l+1}^m, j_{l+2}^m}$ —is also basic.

(v) Up to now, we have shown that the double diagonals $DD_{m+1, j_l^{m+1}, j_{l+1}^{m+1}}$ with $l \in \{1, \dots, l_m - 1\}$ satisfy property (1) of Lemma 6.8. We still have to study the remaining part $DD_{m+1, j_{l_m}^{m+1}, 1}$ of DD_{m+1} .

It is convenient to consider two cases corresponding to the possible values of $j_{l_m}^{m+1}$ from $j_{l_m}^m$.

- *Case $j_{l_m}^{m+1} = j_{l_m}^m$:* Notice (as above in (iv)) that sites $(j_{l_m}^m, 2m - j_{l_m}^m + 2)$ and $(j_{l_m}^m, 2m - j_{l_m}^m + 2)$ —being the leftmost sites of $DD_{m+1, j_{l_m}^{m+1}, j_{l_m}^{m+1}}$ —have state L.

Using the induction hypothesis over m , $DD_{m, j_{l_m}^m, 1}$ is (according to property (2) of the Lemma 6.8 at step m) a 4- (or 5- or 6-) end. As illustrated by Fig. 18(d), the state-values of sites in $DD_{m+1, j_{l_m}^{m+1}, 1}$ are completely determined and this truncated double diagonal is also a 4- (or 5- or 6-) end.

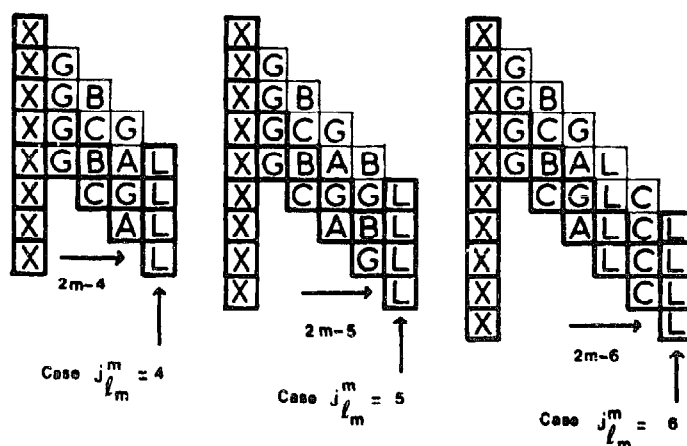


Fig. 18(d).

Note: In this case, we always have $l_{m+1} = l_m$ (which agrees with paragraph (ii)).

– *Case $j_{l_m}^{m+1} = j_{l_m}^m + 1$:* Notice (as above in (iv)) that sites $(j_{l_m}^m + 1, 2m - j_{l_m}^m + 1)$ and $(j_{l_m}^m + 1, 2m - j_{l_m}^m + 2)$ —being the leftmost sites of $DD_{m+1, j_{l_m}^{m+1}, j_{l_m}^{m+1}}$ —have state L.

Using the induction hypothesis over m , $DD_{m, j_{l_m}^m, 1}$ is a 4- (or 5- or 6-) end.

As illustrated by Fig. 18(e) the state values of sites in $DD_{m+1, j_{l_m}^{m+1}, 1}$ are completely determined.

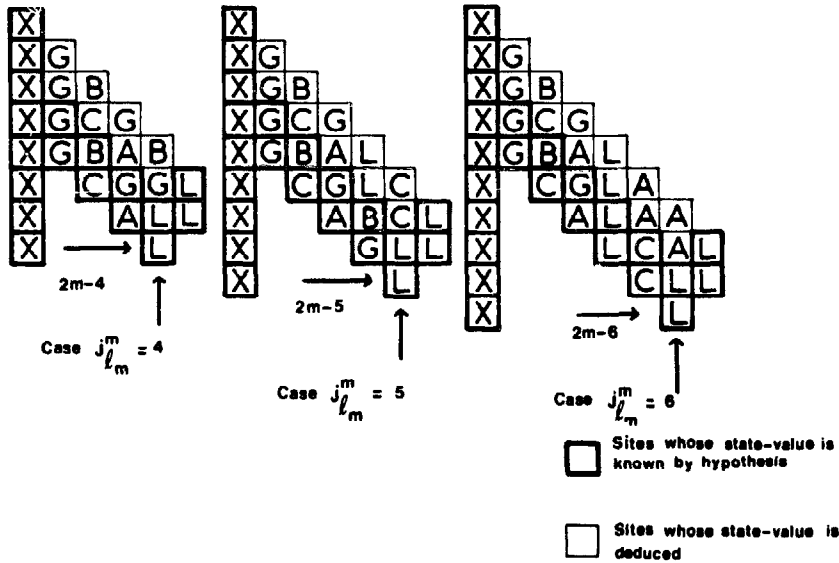


Fig. 18(e).

It is trivial to check that:

- if $DD_{m, j_{l_m}^m, 1}$ is a 4-end (respectively 5-end), then $DD_{m+1, j_{l_m}^{m+1}, 1}$ is a 5-end (respectively 6-end); in this case we have $l_{m+1} = l_m$ which agrees with paragraph (ii).
- if $DD_{m, j_{l_m}^m, 1}$ is a 6-end, then
 - * $DD_{m+1, 4, 1}$ is a 4-end,
 - * $DD_{m+1, 7, 4}$ is A-basic;
 in this case we have $j_{l_m}^m = 6$, $j_{l_m}^{m+1} = 7$, $j_{l_{m+1}}^{m+1} = 4$ and $l_{m+1} = l_m + 1$ which agrees with paragraph (ii).

This completes the proof of Lemma 6.8. $\square$

The following notion is quite convenient.

6.9. Definition. The sets of sites U and V included in $\{1, \dots, n\} \times \{1, \dots, 2n - 1\}$ are n -equivalent if there exists a translation $T: \mathbb{N}^2 \rightarrow \mathbb{N}^2$ such that

- $T(U) = V$;
- for every (K, t) in U , sites (K, t) and $T(K, t)$ have the same state in the state diagram of an initial line of n machines.

We shall use the following consequences of the above proof.

- 6.10. Lemma.** (1) If $j_l^m = j_l^{m+1}$, then the translation $(K, t) \rightarrow (K, t+2)$ maps $DD_{m,j_l^m,1}$ onto $DD_{m+1,j_l^{m+1},1}$ and respects the state values.
- (2) Let $1 < l < l_{m_0}$; the function $m \rightarrow j_l^m - j_{l+1}^m$ with domain $\{m_0, \dots, n-1\}$ is not decreasing.

Proof. (1): From the proof of Lemma 6.8 we see that the condition $j_l^m = j_l^{m+1}$ implies:

- $l_{m+1} = l_m$;
- for s in $\{l, \dots, l_m - 1\}$, the set $DD_{m+1,j_s^{m+1},j_{s+1}^{m+1}}$ and its state values are obtained from DD_{m,j_s^m,j_{s+1}^m} (and its state values) via the translation $(K, t) \rightarrow (K, t+2)$;
- The set $DD_{m+1,j_l^{m+1},1}$ and its state values are also obtained via this translation from $DD_{m,j_l^m,1}$.

(2): This is a trivial consequence of the proof of Lemma 6.8: in fact, $(j_l^{m+1} - j_{l+1}^{m+1}) - (j_l^m - j_{l+1}^m)$ is 0 or 1. $\square$

The next lemma states an essential property of the scheme, i.e., the distribution of states below the reflection. It will be basic to prove that the remaining portion of the initial line after the break at $\frac{2}{3}$ evolves—for the part during and after the reflection—as an initial line of length $\frac{2}{3}n$ (cf. paragraph 6.1).

6.11. Lemma. Let $12 \leq m < n$ be of the form $m = 3p + i$ with $p \geq 4$ and $i \in \{0, 1, 2\}$.

- (1) $DD_{m,m+1,2p+i}$ is A-(respectively, B-, C-) basic if $i = 0$ (respectively 1, 2); also $j_2^m = 2p + i$.
- (2) $DD_{m,2p+i,1}$ is n -equivalent to $DD_{2p+i-1} = DD_{2p+i-1,2p+i,1}$.

Proof. We argue by induction on the integer m . The case $m = 12$ is easily checked (cf. Fig. 16). Suppose now that properties (1) and (2) hold for m .

(i) From Fact 6.1 we see that sites $(m+1, m)$ and $(m+1, m-1)$ have state L. Since $DD_{m,m+1,2p+i}$ is basic, its leftmost sites $(2p+i, 2m-2p-i)$ and $(2p+i, 2m-2p-i+1)$ have state L. Due to the decomposition of DD_m in basic and end DD's (Lemma 6.8), their neighbours $(2p+i-1, 2m-2p-i+1)$ and $(2p+i-1, 2m-2p-i+2)$ have states different from L. This shows that the hypothesis of Lemma 6.6, including condition (5), are satisfied by $DD_{m,m+1,2p+i}$. Applying Lemma 6.6, we are led to the following three cases:

- **Case $i = 0$:** By the induction hypothesis, $DD_{m,m+1,2p+i}$ is A-basic and Lemma 6.6 shows that $DD_{m+1,m+2,2p+i+1}$ is B-basic. Noticing that the equality $m+1 = 3q+j$ with $j \in \{0, 1, 2\}$ implies $q = p$ and $j = i+1 = 1$ so that $2q+j = 2p+i+1$, we see that the first part of property (1) holds for $m+1$.
- **Case $i = 1$:** Similar.
- **Case $i = 2$:** By the induction hypothesis $DD_{m,m+1,2p+i}$ is C-basic and Lemma 6.6 shows that $DD_{m+1,m+2,2p+i}$ is A-basic. Noticing that the equality $m+1 = 3q+j$ with $j \in \{0, 1, 2\}$ implies $q = p+1$ and $j = 0$ so that $2q+j = 2p+2 = 2p+i$, we see that the first part of property (1) holds for $m+1$.

(ii) In the three previous cases, the basic character of $DD_{m+1,m+2,2q+j}$ implies that $j_2^{m+1} = 2q+j$, where $m+1 = 3q+j$ with $j \in \{0, 1, 2\}$. We consider two cases according to the value of j_2^{m+1} from j_2^m .

- *Case $j_2^{m+1} = j_2^m$* : Lemma 6.10(1) and the induction hypothesis prove (2) of this lemma for $m+1$.

- *Case $j_2^{m+1} = j_2^m + 1$* : By induction hypothesis $DD_{m,2p+i,1}$ is equivalent to $DD_{2p+i-1,2p+i,1}$. Sites $(2p+i+1, 2p+i)$ and $(2p+i+1, 2p+i-1)$ have state L by Fact 6.1. Sites $(2p+i+1, 2m-2p-i+1)$ and $(2p+i+1, 2m-2p-i+2)$ have state L because $j_2^{m+1} = j_2^m + 1 = 2p+i+1$ and these sites are the leftmost ones of $DD_{m+1,m+2,j_2^{m+1}}$. Thus, the two sets

$$U = DD_{2p+i-1,2p+i,1} \cup \{(2p+i+1, 2p+i), (2p+i+1, 2p+i-1)\},$$

$$V = DD_{m,2p+i,1} \cup \{(2p+i+1, 2m-2p-i+2), (2p+i+1, 2m-2p-i+1)\}$$

are n -equivalent via the vertical translation of vector $(0, 2m-2(2p+i-1)) = (0, 2p+2)$.

Notice that the state-values of $DD_{m+1,2p+i+1,1}$ (respectively $DD_{2p+i,2p+i+1,1}$) are fully determined by those of V (respectively U) (see Fig. 18(f)). We thus deduce that $DD_{m+1,2p+i+1,1}$ and $DD_{2p+i,2p+i+1,1}$ are n -equivalent.

Observe that the equality $j_2^{m+1} = j_2^m + 1$ implies that $DD_{m,m+1,j_2^m}$ is A- or B-basic (cf. paragraph (ii) of the proof of Lemma 6.8). Thus, Lemma 6.11(1) shows that if $m = 3p+i$, then $i \neq 2$.

Finally, recall that if $m+1 = 3q+j$ (where $j \in \{0, 1, 2\}$), then $2q+j = 2p+i+1$ (where $m = 3p+i$ with $i \in \{0, 1\}$).

Thus the n -equivalent preceding sets are exactly those of Lemma 6.11(2) for $m+1$. $\square$

6.3. The two next lemmas describe some properties of the reflection of the initial wave. The first one considers the emission and transmission by the right-end soldier of the reflected initial wave.

6.12. Lemma *Let $n \geq 25$, $n = 3p+i$ with $i \in \{1, 2, 3\}$. Let Z and U be the following sets:*

$$Z = \{(l, 2n-l+i+z) \mid l \in \{j_2^{n-1}+5, \dots, n\}, z \in \{-1, 0, 1, 2, 3\}\},$$

$$U = \{(l, 2n-l+u) \mid l \in \{j_2^{n-1}+5, \dots, n\}, u \in \{0, \dots, 6\}\} \quad (\text{cf. Fig. 19(b)}).$$

These sets are nonempty and it holds that

(1) *the distribution of states on Z is as indicated in Fig. 19(a), i.e., every vertical section has states CBLLL from bottom to top;*

(2) *the distribution of states on U completes that of Z as indicated in Fig. 19(a):*

Case $i = 1$: $U \setminus Z$ consists of two L-valued diagonals above Z .

Case $i = 2$: $U \setminus Z$ consists of two diagonals, one A-valued below Z and the other L-valued above Z .

Case $i = 3$: $U \setminus Z$ consists of two diagonals below Z . The highest one is A-valued, the other one is B-valued except its rightmost site which has value G.

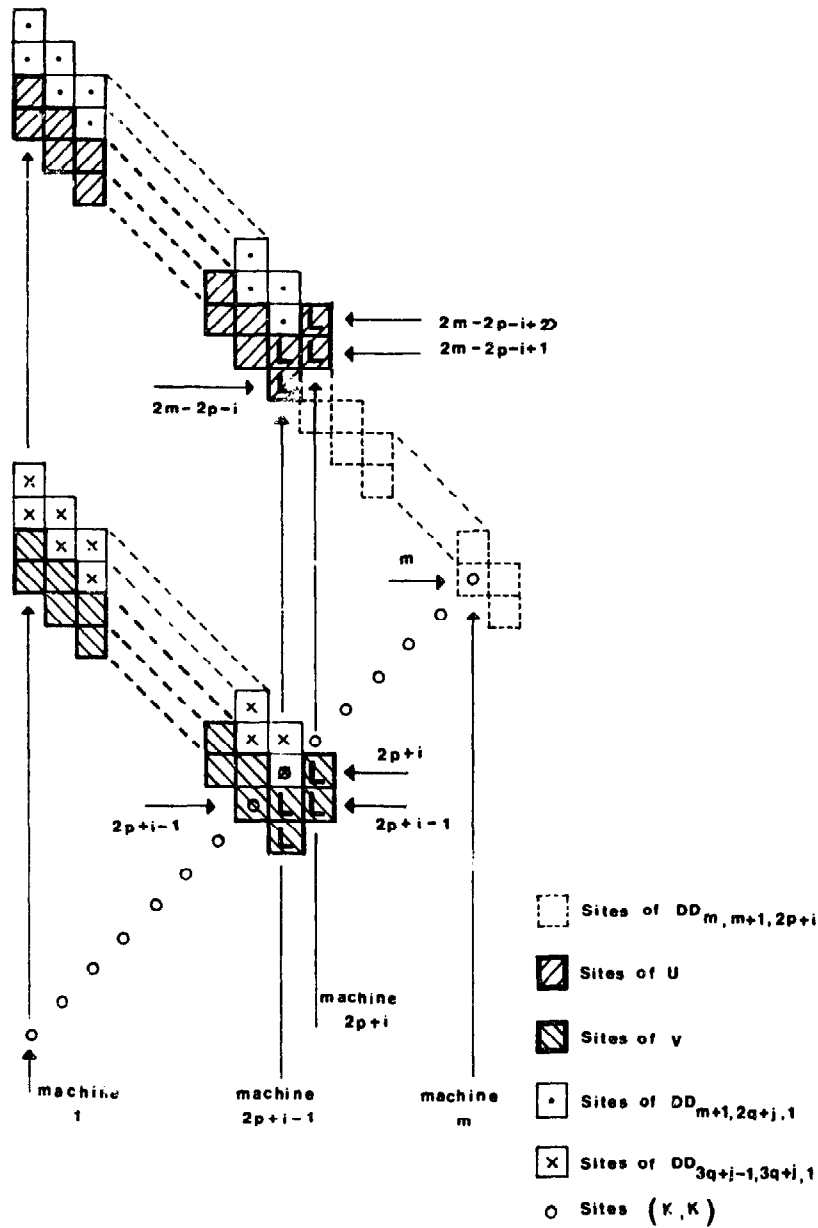


Fig. 18(f).

Proof. When $n = 25$, we have $j_2^{24} = 16$ and the truncated double diagonal $DD_{24,25,16}$ has sites corresponding to ten adjacent machines. Using Lemma 6.10(2), (we deduce that for all $n \geq 25$, $DD_{n-1,n,j_2^{n-1}}$ has at least ten adjacent machines.

As illustrated in Fig. 19(b), $DD_{n-1,n,j_2^{n-1}}$ fully determines the set U . Observe that this set is nonempty since there are least six (indeed ten) adjacent machines involved in $DD_{n-1,n,j_2^{n-1}}$.

Observing that $n-1 = 3p + (i-1)$ with $(i-1) \in \{0, 1, 2\}$, Lemma 6.11—applied to $DD_{n-1,n,j_2^{n-1}}$ —gives the three distributions of states on U (as illustrated in Fig. 19(b)). $\square$

Remark. Lemma 6.12 proves that the delay is really conveyed by the reflected wave as indicated in Section 4.

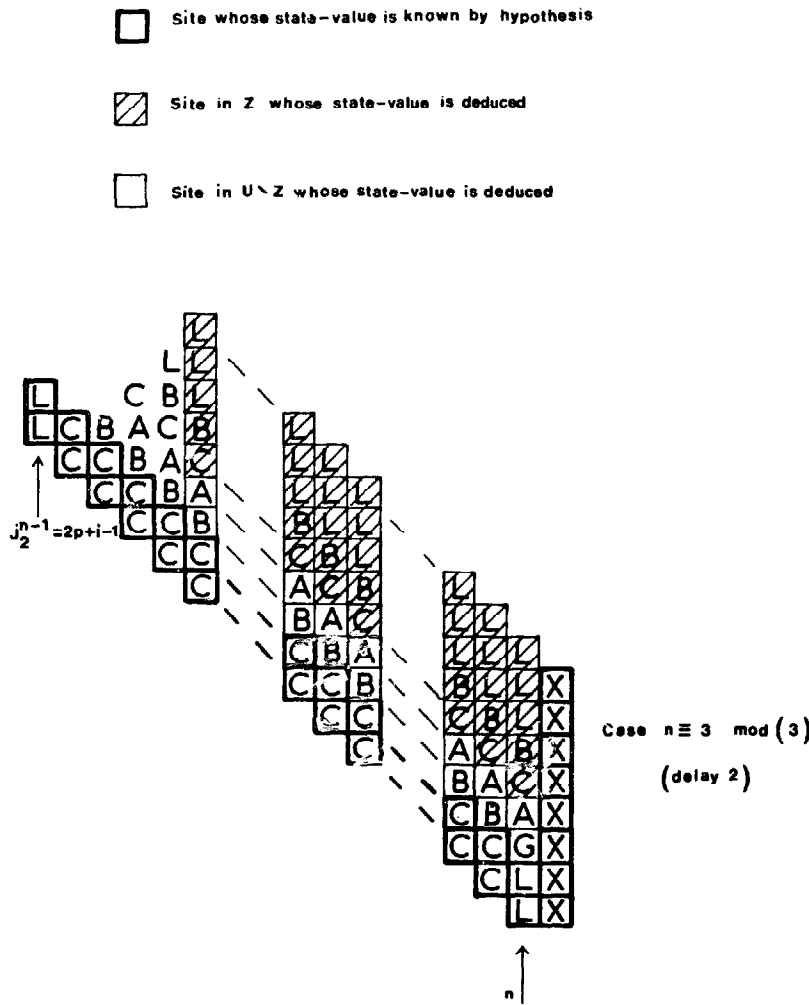


Fig. 19(a). Part 1.

The following lemma considers the crash of the reflection of the initial wave and completes Lemma 6.12.

6.13. Lemma. *Let $n \geq 25$. The distribution of states for sites corresponding to machines $j_2^{n-1} - 2, \dots, j_2^{n-1} + 5$ and to the arrival of the reflection of the initial wave is always that of some of the nine cases indicated in Fig. 19(c).*

Proof. Let V be the set

$$DD_{n-1, j_2^{n-1}+5, j_2^{n-1}-2} \cup \{(j_2^{n-1}+5, 2n-j_2^{n-1}-5+l) \mid l \in \{0, \dots, 6\}\}.$$

The set $\tilde{V}$ is fully determined by V .

We first observe that $j_2^{24} > j_3^{24} + 3$ and $25 = j_1^{24} > j_2^{24} + 6$ (easy check, cf. Fig. 16). Lemma 6.10(2) permits to extend these inequalities: $j_2^{n-1} > j_3^{n-1} + 3$ and $n = j_1^{n-1} > j_2^{n-1} + 6$. This ensures that

$$DD_{n-1, j_2^{n-1}+5, j_2^{n-1}-2} \subset (DD_{n-1, j_2^{n-1}, j_3^{n-1}} \cup DD_{n-1, n, j_2^{n-1}}).$$

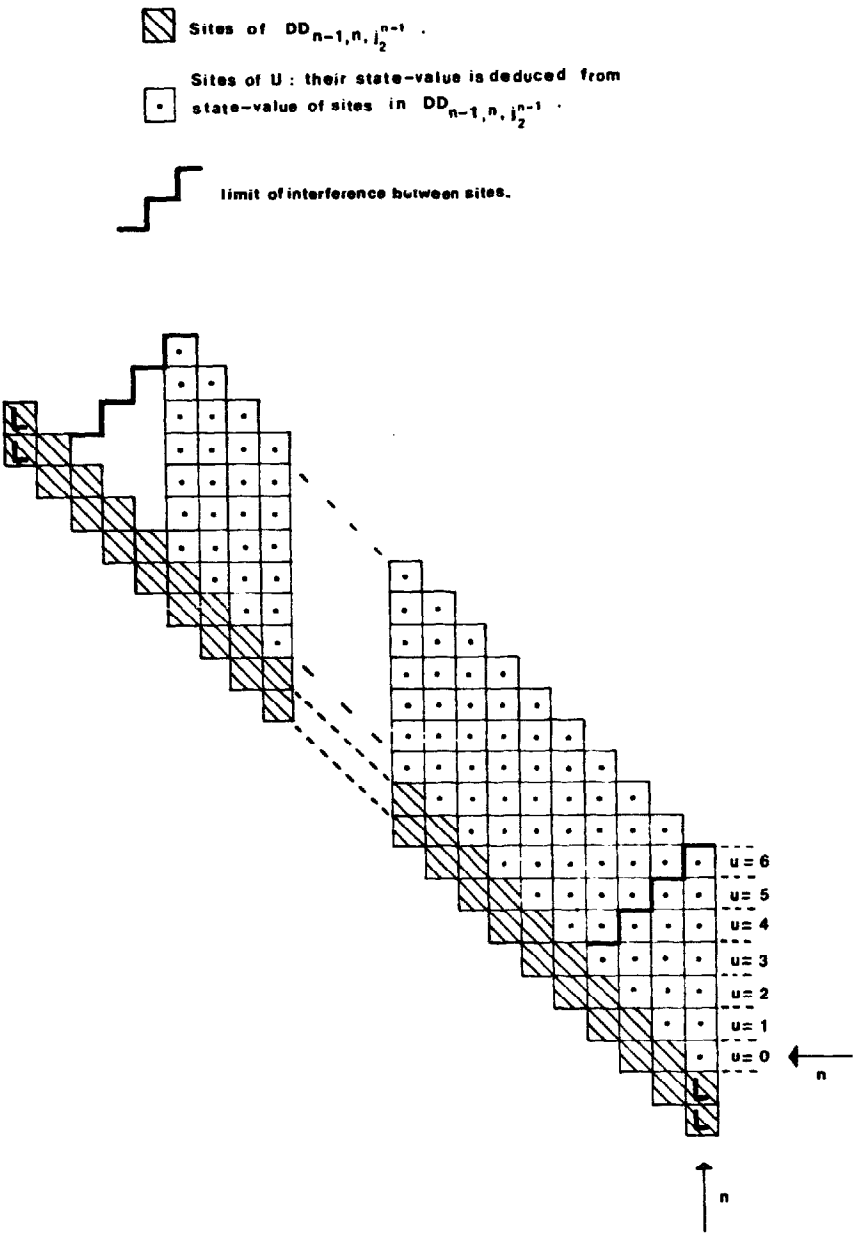


Fig. 19(b).

6.4. In order to compare state diagrams for initial lines of different lengths, it is convenient to introduce the following notation and definition.

6.14. **Notation.** Let Y be a set of sites included in $\{1, \dots, N\} \times \{1, \dots, 2N - 1\}$ and $N \leq n$. By $\langle Y \rangle_n$ we mean the restriction to Y of the state diagram of the evolution of an initial line of length n . Also $\langle K, t \rangle_n$ denotes $\langle \{K, t\} \rangle_n$.

6.15. **Definition.** Let Y and Z be two sets of sites. By $\langle Y \rangle_n \sim \langle Z \rangle_m$ we mean

- $\langle Y \rangle_n$ and $\langle Z \rangle_m$ are both defined, and
- there exists a translation $T: (K, t) \rightarrow (K - a, t - b)$ such that $T(Y) = Z$ and T respects state values (i.e., $\langle T(K, t) \rangle_m = \langle K, t \rangle_n$ for (K, t) in Y).

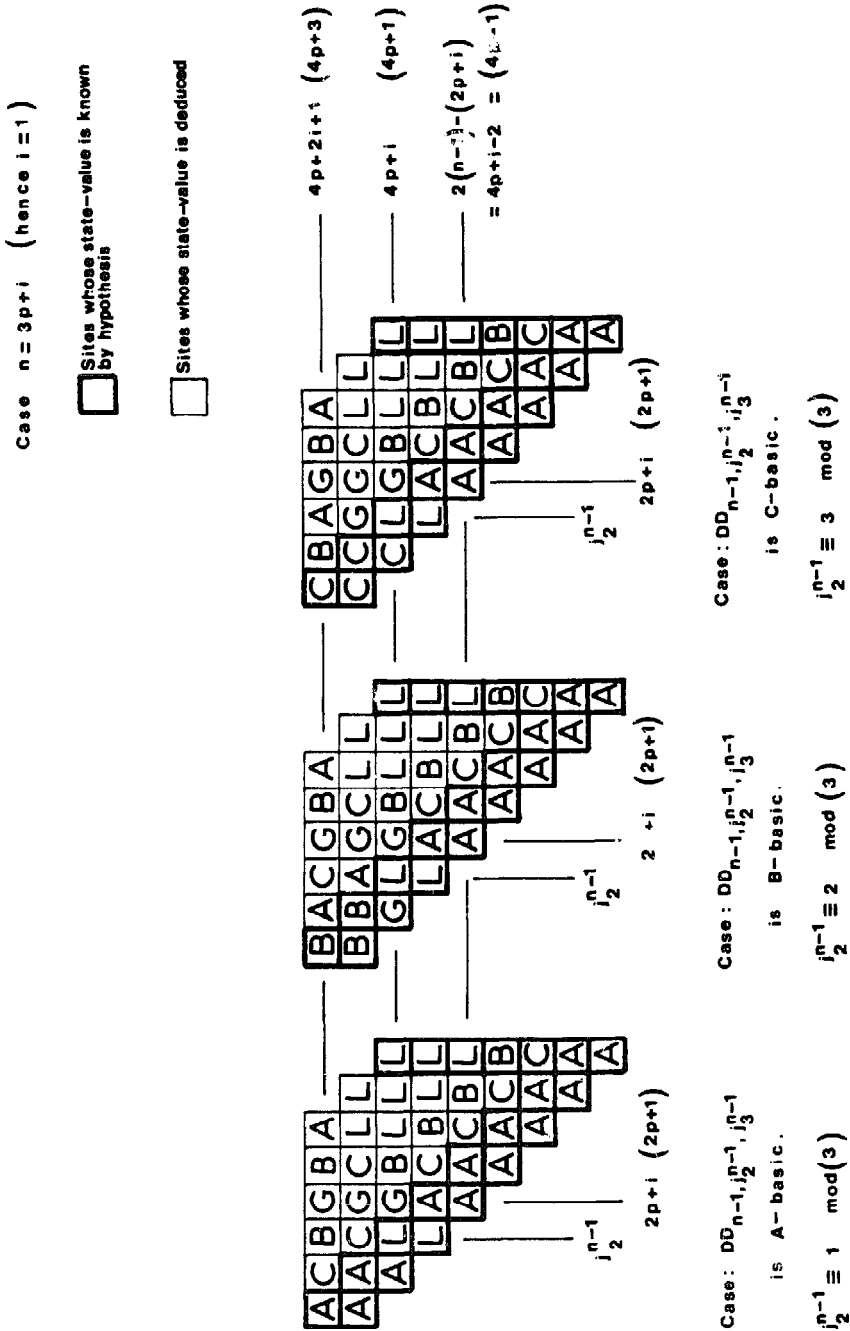
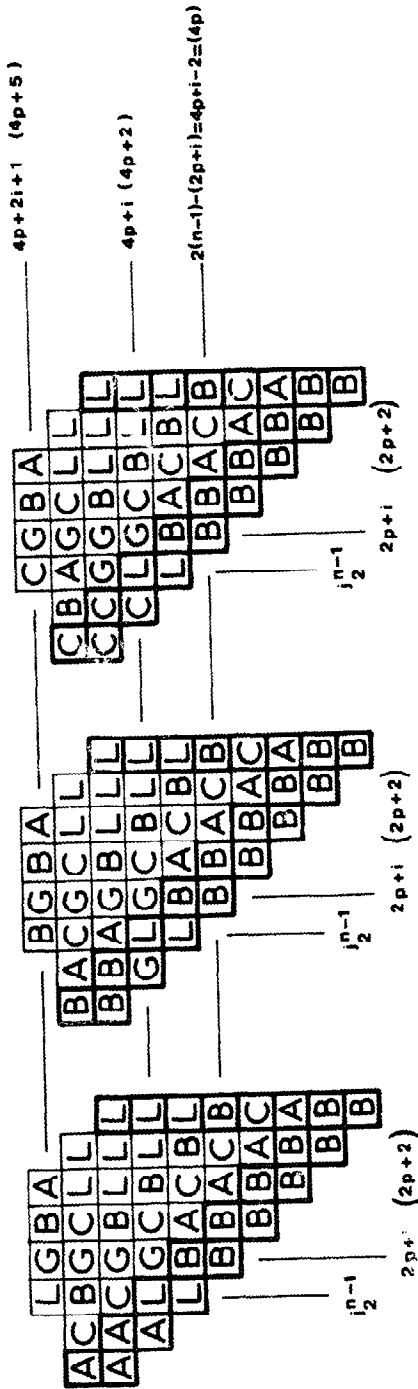


Fig. 19(c). Part 1.

Case $n = 3p + 2$ (hence $i = 2$)

☐ Sites whose state-value is known by hypothesis

☐ Sites whose state-value is deduced



Case: $DD_{n-1, j_2^{n-1}, j_3^{n-1}}$

is C-basic.

$$j_2^{n-1} \equiv 3 \pmod{3}$$

Case: $DD_{n-1, j_2^{n-1}, j_3^{n-1}}$

is B-basic.

$$j_2^{n-1} \equiv 2 \pmod{3}$$

Case: $DD_{n-1, j_2^{n-1}, j_3^{n-1}}$

is A-basic.

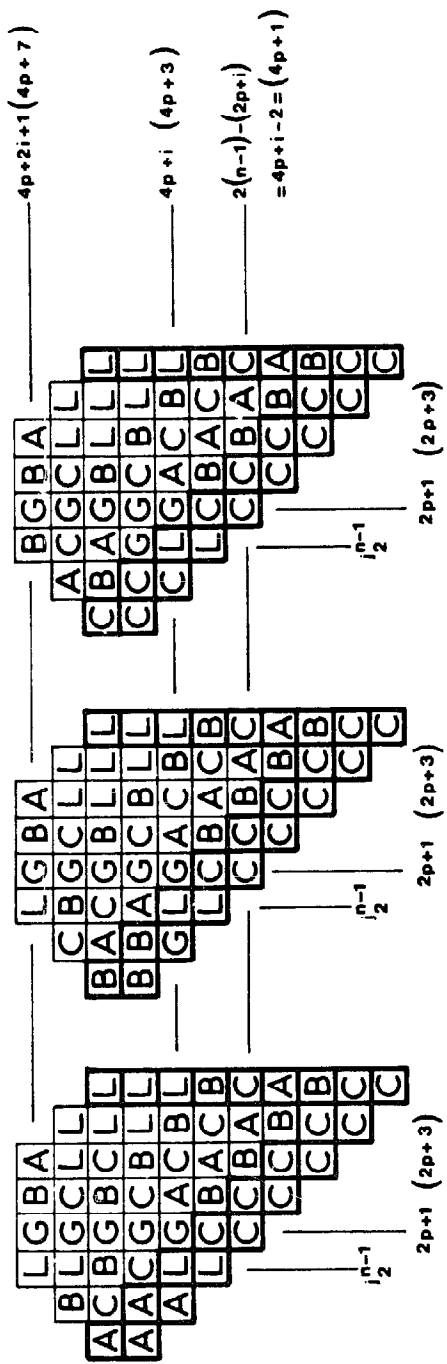
$$j_2^{n-1} \equiv 1 \pmod{3}$$

Fig. 19(c). Part 2.

Case $n = 3p + 3$ (hence $i \equiv 3$)

☐ Sites whose state-value is known by hypothesis

☐ Sites whose state-value is deduced



Case: $DD_{n-1, j_2^{n-1}, j_3^{n-1}}$ is A-basic $j_2^{n-1} \equiv 1 \pmod{3}$

Case: $DD_{n-1, j_2^{n-1}, j_3^{n-1}}$ is B-basic $j_2^{n-1} \equiv 2 \pmod{3}$

Case: $DD_{n-1, j_2^{n-1}, j_3^{n-1}}$ is C-basic $j_2^{n-1} \equiv 3 \pmod{3}$

Fig. 19(c). Part 3.

Remark. In case $n = m$, then $\langle Y \rangle_n \sim \langle Z \rangle_m$ if and only if Y and Z are n -equivalent in the sense of Definition 6.9.

6.16. Notation. $H'_{K,j}$ is the following portion of a constant-time line:

$$\{(l, t) \mid K \leq l \leq j\}.$$

6.17. Lemma. Let $n \geq 25$, $n = 3p + i$ with $i \in \{1, 2, 3\}$, then

$$\langle H'_{2p+i, 3p+i} \rangle_{3p+i} \sim \langle H^3_{1, p+1} \rangle_{p+1}$$

for $t = (2n - 1) - [2(p + 1) - 1 - 3] = 4p + 2i + 1$ (i.e., the distribution of states at time 3 for an initial line of $p + 1$ machines is that at time t for the $p + 1$ rightmost machines of an initial line of n machines).

Proof. It is clear that the distribution of states on $H^3_{1, p+1}$ for an initial line of length $p + 1$ is GBALL...LL.

Using Lemma 6.13 for a line of n machines, it is easy to check that, in the nine cases of Fig. 19(c),

- sites $(2p + i, t)$, $(2p + i + 1, t)$, $(2p + i + 2, t)$ are in respective states GBA;
- sites $(2p + i + 2, t - 1)$, $(2p + i + 2, t - 2)$, $(2p + i + 3, t - 2)$, $(2p + i + 3, t - 3)$ are in state L.

By Lemma 6.11, $j_2^{n-1} = 2p + i - 1$; thus, Lemma 6.12 ensures that all sites in the set

$$W = \{(l, 2n - l + i + 1), (l, 2n - l + i + 2) \mid l \in \{2p + i + 4, \dots, n\}\}$$

have state L (for an initial line of n machines).

Observing (cf. Fig. 20(a)) that the set

$$W \cup \{(2p + i + 2, t - 1), (2p + i + 2, t - 2), (2p + i + 3, t - 2), (2p + i + 3, t - 3)\}$$

fully determines a set containing $H'_{2p+i+3, 3p+i}$, we deduce that all sites in $H'_{2p+i+3, 3p+i}$ have state L (for an initial line of n machines).

This proves that the states of $\langle H'_{2p+i, 3p+i} \rangle_{3p+i}$ are GBALL...LL.

Taking $T: (h, \theta) \rightarrow (h - 2p - i + 1, \theta - t + 3)$, we see that $\langle H'_{2p+i, 3p+i} \rangle_{3p+i} \sim \langle H^3_{1, p+1} \rangle_{p+1}$. $\square$

6.18. Notation. $LDD'_{m,j}$ is the following set of sites (it is the union of an initial truncated double diagonal and a triangle) (cf. Fig. 20(b)):

$$LDD'_{m,j} = DD_{m, j, 1} \cup \{(h, \theta) \mid h \leq j \text{ and } \theta \leq t \text{ and } h + \theta \geq 2m + 1\}.$$

6.19. Lemma. Let $n \geq 25$, $n = 3p + i$ with $i \in \{1, 2, 3\}$, then

$$\langle LDD'_{n-1, 2p+i-1} \rangle_n \sim \langle LDD'_{2p+i-2, 2p+i-1} \rangle_{2p+i-1}$$

(i.e., the distribution of states at time $2p + i - 1 + b$ (with $b \in \{0, \dots, i\}$) of the b rightmost machines of an initial wave of length $2p + i - 1$ is that at time $4p + i + 1 + b$

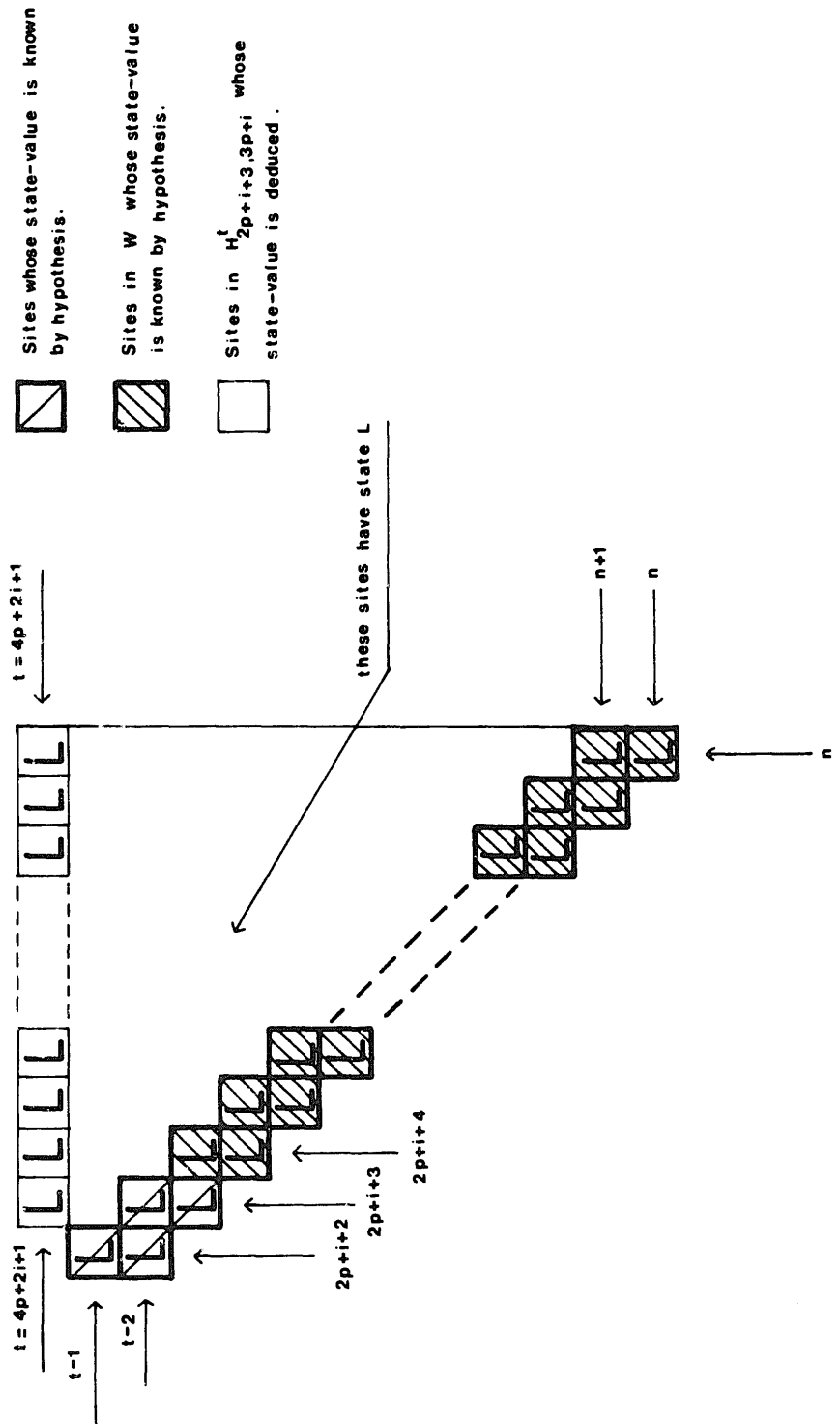


Fig. 20(a).

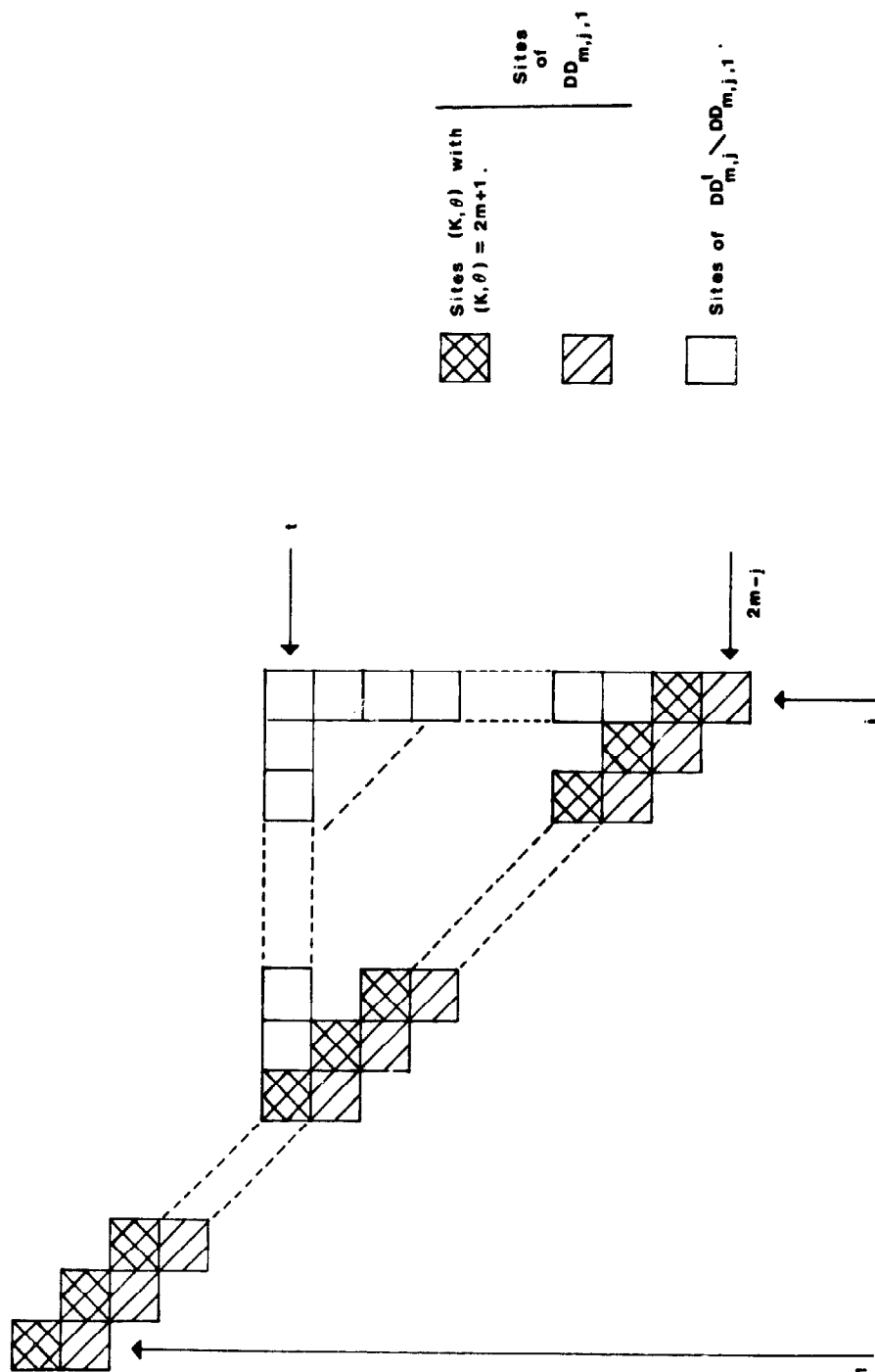


Fig. 20(b).

(with $b \in \{0, \dots, i\}$) of the b left neighbours of the machine $2p+i$ of an initial line of n machines.

Proof. (i) Observing that if $n = 3p+i$ with $n \in \{1, 2, 3\}$, then $n-1 = 3p+i-1$ with $i-1 \in \{0, 1, 2\}$, Lemma 6.11 ensures that

$$\langle DD_{n-1, 2p+i-1, 1} \rangle_n \sim \langle DD_{2p+i-2, 2p+i-1, 1} \rangle_n.$$

Let I be the set

$$\{(h, h-l) \mid 0 < l \leq h-1 \text{ and } h \leq 2p+i\};$$

then $DD_{2p+i-2, 2p+i-1, 1}$ is fully determined by $I \cup \{(1, 1)\}$.

By Fact 6.1 we have $\langle I \rangle_n = \langle I \rangle_{2p+i-1}$ (all sites have state L). Since $\langle 1, 1 \rangle_n = \langle 1, 1 \rangle_{2p+i-1} = G$, we have

$$\langle DD_{2p+i-2, 2p+i-1, 1} \rangle_n \sim \langle DD_{2p+i-2, 2p+i-1, 1} \rangle_{2p+i-1}.$$

Thus, $\langle DD_{n-1, 2p+i-1, 1} \rangle_n \sim \langle DD_{2p+i-2, 2p+i-1, 1} \rangle_{2p+i-1}$.

(ii) States $\langle 2p+i-1, 4p+i+1+b \rangle_n$ with $b \in \{0, \dots, i\}$ are indicated by Lemma 6.13. These states depend on the remainder of $2p+i-1$ modulo 3; they are given in Fig. 19(c).

States $\langle 2p+i-1, 2p+i-1+b \rangle_{2p+i-1}$ with $b \in \{0, \dots, i\}$ are fully determined by the states of DD_{2p+i-2} for an initial line of $2p+i-1$ machines; they are given in Fig. 19(a) (where n is replaced by $2p+i+1$). These states depend also on the remainder of $2p+i-1$ modulo 3.

Whatever may be this remainder, one easily checks in Figs. 19(c) and 19(a), via the translation $T: (h, t) \rightarrow (h, t-2p-2)$, that

$$\begin{aligned} & \langle \{(2p+i-1, 4p+i+1+b) \mid b \in \{0, \dots, i\}\} \rangle_n \\ & \sim \langle \{(2p+i-1, 2p+i-1+b), b \in \{0, \dots, i\}\} \rangle_{\sim_{\bar{p}, i-1}}. \end{aligned}$$

(iii) Observing that $LDD_{n-1, 2p+i-1}^{4p+2i+1}$ and $LDD_{2p+i-2, 2p+i-1}^{2p+2i-1}$ are fully determined by

$$DD_{n-1, 2p+i-1, 1} \cup \{(2p+i-1, 4p+i+1+b) \mid 0 \leq b \leq i\}$$

and

$$DD_{2p+i-2, 2p+i-1, 1} \cup \{(2p+i-1, 2p+i-1+b) \mid 0 \leq b \leq i\},$$

the two preceding points show that

$$\langle LDD_{n-1, 2p+i-1}^{4p+2i+1} \rangle_n \sim \langle LDD_{2p+i-2, 2p+i-1}^{2p+2i-1} \rangle_{2p+i-1}. \quad \square$$

6.5. Let $n = 3p+i$ with $i \in \{1, 2, 3\}$ and $n \geq 25$; we observe that

- $LDD_{n-1, 2p+i-1}^{4p+2i+1} \cup \{(2p+i, \theta) \mid 4p+2i+1 \leq \theta \leq 2n-1\}$ fully determines $\{(K, 2n-1) \mid 1 \leq K \leq 2p+i-1\}$ (i.e., the (wanted) synchronization step of the $2p+i-1$ leftmost machines of the initial line at time $2n-1$);
- $H_{2p+i, 3p+i}^{4p+2i+1} \cup \{(2p+i, \theta) \mid 4p+2i+1 \leq \theta \leq 2n-1\}$ fully determines $\{(K, 2n-1) \mid 2p+i \leq K \leq 3p+i\}$ (i.e., the (wanted) synchronization step of the $p+1$ rightmost machines of the initial line at time $2n-1$).

Lemmas 6.17 and 6.19 show that the distribution of states on $LDD_{n-1, 2p+i-1}^{4p+2i+1}$ and $H_{2p+i, 3p+i}^{4p+2i+1}$ are those on adequate subsets of the state diagrams of initial lines having lengths strictly shorter than n .

In order to reduce the synchronization of an initial line of n machines to that of shorter initial lines, we prove that (Σ) :

- machine $2p+i$ stays in state G from time $4p+2i+1$ up to time $2n-2$;
- During the same time interval, the states U, V of machines $2p+i-2$ and $2p+i-1$ are such that the rules for environments (U, V, X) and (U, V, G) give the same result.

6.20. Lemma. *Let $n = 3p+i$ with $i \in \{1, 2, 3\}$ and $n \geq 25$. For every time θ in $\{1, \dots, 4p+2i+1\}$ $(\langle n-1, \theta \rangle, \langle n, \theta \rangle)$ is an element of the following set $\Lambda_1 = \{(C, L), (A, L), (A, C), (C, B), (B, L), (L, L), (G, L), (C, G), (B, A)\}$.*

Note. This proves that up to time $4p+2i+1$, the environments GGX and AAX to not occur.

Proof of Lemma 6.20. From Fact 6.1 we know that these two states are (L, L) up to time $n-2$. From Lemma 6.12 one checks that these two states are in Λ_1 from time $n-1$ up to time $n+6$.

From Lemma 6.13 one checks that these two states are (L, L) from time $n+7$ up to time $4p+2i+1$. $\square$

We cannot prove (Σ) directly. We proceed by an examination of successive horizontal lines.

6.21. Lemma. *Let $n \geq 25$ and $n = 3p+i$ with $i \in \{1, 2, 3\}$. Suppose that the following condition:*

$$\begin{aligned} &\text{for all } \theta, 1 \leq \theta \leq 2m-3: (\langle m-1, \theta \rangle_m, \langle m, \theta \rangle_m) \in \Lambda_2 \\ &\text{with } \Lambda_2 = \Lambda_1 \cup \{(B, C), (B, G), (G, C), (G, B), (G, A)\} \end{aligned} \quad (6)$$

holds for all m such that $3 \leq m \leq n$. Then

- (1) For $t \in \{0, \dots, 2p-3\}$,
 - (a) $\langle H_{2p+i, 3p+i}^{4p+2i+1+t} \rangle_{3p+i} \sim \langle H_{1, p+1}^{3+t} \rangle_{p+1}$;
 - (b) $\langle H_{2p-2-t, 2p+i-1}^{4p+2i+1+t} \rangle_{3p+i} \sim \langle H_{2p-2-t, 2p+i-1}^{2p+2i-1+t} \rangle_{2p+i-1}$;
- (2) Condition (6) holds for $m = n$.

Proof. (i) We prove statement (1) by induction on t . Observing that $H_{2p-2, 2p+i-1}^{4p+2i+1}$ is included in $LDD_{n-1, 2p+i-1}^{4p+2i+1}$ we see that the case $t=0$ is answered by Lemmas 6.17 and 6.19.

We now suppose that statement (1) is true for some $t \leq 2p-4$ and we prove (1) for $t+1$ in (ii) to (v) below (cf. Fig. 21).

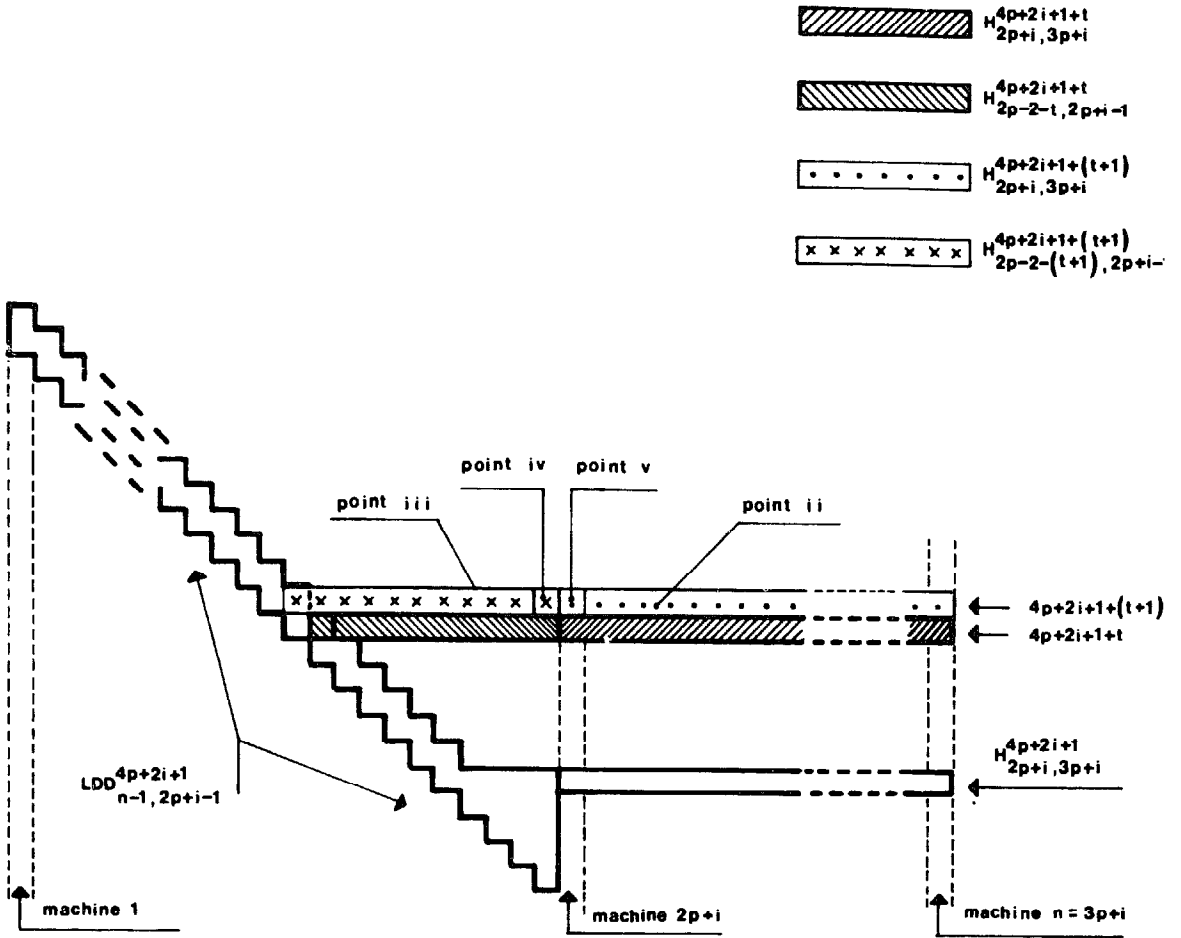


Fig. 21.

(ii) Since $H_{2p+i+1, 3p+i}^{4p+2i+1+(t+1)}$ (respectively $H_{2,p+1}^{3+(t+1)}$) is fully determined by $H_{2p+i, 3p+i}^{4p+2i+1+t}$ (respectively $H_{1,p+1}^{3+t}$), Lemma 6.21(a) shows that

$$\langle H_{2p+i+1, 3p+i}^{4p+2i+1+(t+1)} \rangle_n \sim \langle H_{2,p+1}^{3+(t+1)} \rangle_{p+1}.$$

(iii) Observe that:

- $H_{2p-2-(t+1), 2p+i-2}^{4p+2i+1+(t+1)}$ (respectively $H_{2p-2-(t+1), 2p+i-2}^{2p+2i-1+(t+1)}$) is fully determined by $H_{2p-2-t, 2p+i-1}^{4p+2i+1+t} \cup LDD_{n-1, 2p+i-1}^{4p+2i+1}$ (respectively $H_{2p-2-t, 2p+i-1}^{2p+2i-1+t} \cup LDD_{2p+i-2, 2p+i-1}^{2p+2i-1}$);
- $\langle H_{2p-2-t, 2p+i-1}^{4p+2i+1+t} \rangle_n \sim \langle H_{2p-2-t, 2p+i-1}^{2p+2i-1+t} \rangle_{2p+i-1}$, by Lemma 6.21(b) for t ;
- $LDD_{n-1, 2p+i-1}^{4p+2i+1} \sim \langle LDD_{2p+i-2, 2p+i-1}^{2p+2i-1} \rangle_{2p+i-1}$, by Lemma 6.19.

Thus, $\langle H_{2p-2-(t+1), 2p+i-2}^{4p+2i+1+(t+1)} \rangle_n \sim \langle H_{2p-2-(t+1), 2p+i-2}^{2p+2i-1+(t+1)} \rangle_{p+1}$.

(iv) Observe that if $3 \leq \theta \leq 2p$, then $\langle 1, \theta \rangle_{p+1} = G$;

- $\theta \geq 14$, then $(1, \theta)$ is a leftmost site of a double diagonal DD_m with $7 \leq m < p+1$ and Lemma 6.8 ensures that $\langle 1, \theta \rangle_{p+1} = G$;
- Cases $3 \leq \theta \leq 13$ are easily checked in Fig. 16.

Now we complete the distribution of states on $H_{2p-2-(t+1), 2p+i-1}^{4p+2i+1+(t+1)}$ by determining the state value of site $(2p+i-1, 4p+2i+1+(t+1))$. Condition (6) shows that

$$(\langle 2p+i-2, 2p+2i-1+t \rangle_{2p+i-1}, \langle 2p+i-1, 2p+2i-1+t \rangle_{2p+i-1}) \in \Lambda_2.$$

Lemma 6.21(b) for t then shows that

$$(\langle 2p+i-2, 4p+2i+1+t \rangle_n, \langle 2p+i-1, 4p+2i+1+t \rangle_n) \in \Lambda_2.$$

It is easy to check from the transition matrices (cf. Fig. 15) that if $(U, V) \in \Lambda_2$, then (U, V, X) and (U, V, G) are environments for which there are transition rules, and these two rules have the same result. Observing that $\langle 2p+i, 4p+2i+1+t \rangle_n$ is $\langle 1, 3+t \rangle_{p+1}$ by Lemma 6.21(1)(a) for t and $\langle 1, 3+t \rangle_{p+1}$ is state G (cf. above), we see that the environments

$$(\langle 2p+i-2, 2p+2i-1+t \rangle_{2p+i-1}, \langle 2p+i-1, 2p+2i-1+t \rangle_{2p+i-1}, X)$$

and

$$(\langle 2p+i-2, 4p+2i+1+t \rangle_n, \langle 2p+i-1, 4p+2i+1+t \rangle_n, G)$$

have the same result, hence

$$\langle 2p+i-1, 4p+2i+1+(t+1) \rangle_n = \langle 2p+i-1, 2p+2i-1+(t+1) \rangle_{2p+i-1}$$

(v) Now we complete the distribution of states on $H_{2p+i, 3p+i}^{4p+2i+1+t+1}$ by determining the state value of site $(2p+i, 4p+2i+1+(t+1))$. Observing that if $t \leq 2p-4$, then $3+t \leq 2p-1$, we note that $\langle 1, 3+t \rangle_{p+1}$ and $\langle 1, t+3+1 \rangle_{p+1}$ are in state G (cf. above).

In an initial line of $p+1$ machines, site $(2, 3+t)$ is on double diagonal DD_s (where s is $\frac{1}{2}(5+t)$ if t is odd $\frac{1}{2}(5+t-1)$ otherwise). Since we have $s \leq p$, site $(2, 3+t)$ is a site for machine 2 on a 4- (or 5- or 6-) end by Lemma 6.8. Thus, $\langle 2, 3+t \rangle_{p+1}$ is state B or C (cases $3 \leq t+3 \leq 13$ are directly checked in Fig. 16). By statement (1)(a) for t , $\langle 2p+i-1, 4p+2i+1+t \rangle_n$ is B or C.

Since $n \geq 25$, we have $p \geq 2$ and $2p+i-1 \geq 3$. Thus we can apply condition (6) to $m = 2p+i-1$ which shows that

$$(\langle 2p+i-2, 2p+2i-1+t \rangle_{2p+i-1}, \langle 2p+i-1, 2p+2i-1+t \rangle_{2p+i-1}) \in \Lambda_2.$$

hence that $\langle 2p+i-1, 2p+2i-1+t \rangle_{2p+i-1}$ is different from state F.

Thus, by (1)(b) for t , $\langle 2p+i-1, 4p+2i+1+t \rangle_n$ is different from state F. Observing—on the transition matrices of Fig. 15—that all environments (U, G, B) and (U, G, C) (where U is different from F) give state G, we conclude that $\langle 2p+i, 4p+2i+1+(t+1) \rangle_n$ is state G. And this completes the proof of Lemma 6.21(1) for $t+1$.

(vi) Now we consider statement (2). Lemma 6.20 shows that condition (6) for n holds from time 1 up to time $4p+2i+1$. By (1), $(\langle n-1, 4p+2i+1+t \rangle_n, \langle n, 4p+2i+1+t \rangle_n)$ is $(\langle p, 3+t \rangle_{p+1}, \langle p+1, 3+t \rangle_{p+1})$ for $0 \leq t \leq 2p-3$.

Thus, condition (6) for $p+1 < 3p+i = n$ ensures us that $(\langle n-1, 4p+2i+1+t \rangle_n, \langle n, 4p+2i+1+t \rangle_n)$ is in Λ_2 for $1 \leq 3+t \leq 2(p+1)-3$, i.e., for $4p+2i+1 \leq 4p+2i+1+t \leq 4p+2i+1+2p-4 = 2n-3$.

This proves that condition (6) holds for n . $\square$

6.6. Now we can complete the proof of correctness.

6.22. Theorem. Let $\mathfrak{A}$ be the automaton whose transition rules are indicated in Fig. 15; $\mathfrak{A}$ is a minimal time solution of the firing squad synchronization problem.

Proof. We prove by induction on the length n of the initial line that

- (1) $\mathfrak{A}$ synchronizes an initial line of length n in time $2n - 1$;
- (2) $\mathfrak{A}$ G-synchronizes an initial line of length n with $n \geq 3$ (i.e., $\langle K, 2n - 2 \rangle_n = G$ for $K \in \{1, \dots, n\}$);
- (3) condition (6) holds for n with $n \geq 3$.

We easily check the following facts:

- $\mathfrak{A}$ is a minimal-time solution of the firing squad for initial lines strictly shorter than 25;
- for $3 \leq n \leq 24$, $\mathfrak{A}$ G-synchronizes an initial line of length n ;
- condition (6) holds for n with $3 \leq n \leq 24$.

We now suppose that facts (1)–(3) are true for $m \leq n - 1$, and we prove them for n with $n \geq 25$.

Induction hypothesis (3) for $m \leq n - 1$ shows that the hypothesis of Lemma 6.21 holds for n . Thus, Lemma 6.21(1)(a) in the case of $t = 2p - 3$ shows that $\langle H_{2p+i, 3p+i}^{2n-2} \rangle_n \sim \langle H_{1, p+1}^{2p} \rangle_{p+1}$.

Since $p + 1 < 3p + i = n$, induction hypothesis (2) for $p + 1$ shows that all sites $(K, 2p)$ (with $1 \leq K \leq p + 1$) are in state G if the initial line is of length $p + 1$. Thus, all sites $(K, 2n - 2)$ with $K \in \{2p + i, \dots, 3p + i\}$ are in state G if the initial line has length n .

Lemma 6.21(1)(b) in the case of $t = 2p - 3$ shows that $\langle H_{1, 2p+i-1}^{2n-2} \rangle_n \sim \langle H_{1, 2p+i-1}^{4p+2i-4} \rangle_{2p+i-1}$.

Since $2p + i + 1 < n$ ($n = 3p + i$ with $p \geq 7$ and $i \in \{1, 2, 3\}$), induction hypothesis (2) for $2p + i + 1$ shows that all sites $(K, 2(2p + i - 1) - 2)$, with $1 \leq K \leq 2p + i - 1$, are in state G if the initial line has length $2p + i - 1$. Thus, all sites $(K, 2n - 2)$ —with $1 \leq K \leq 2p + i - 1$ —are in state G if the initial line is of length n .

In this way we have proved that all sites $(K, 2n - 2)$ with $K \in \{1, \dots, n\}$ are in state G if the initial line is of length n . This proves point (2) for n .

From this G-synchronization at time $2n - 2$, we get—using rules $(G, G, G) \rightarrow F$, $(X, G, G) \rightarrow F$, $(G, G, X) \rightarrow F$ —the F-synchronization at time $2n - 1$ of the initial line of length n .

Finally, condition (6) for n is a trivial consequence of Lemma 6.21 (and the induction hypothesis). $\square$

7. Conclusion

In [1], Balzer has shown that

- no minimal time solution exists with four states;
- no minimal time solution satisfying extra conditions exists with five states.

However, the solution presented here does not satisfy Balzer's four extra conditions:

- in particular, his extra conditions 1 (the stability of state G) and 4 (rules $(G, V, G) \rightarrow G$ for $V \neq G$) are violated,
- the very idea of our solution is *not to be* an 'image solution' (his condition 2);
- the only condition satisfied is condition 3 (the fire is introduced only by environments GGG, XGG, GGX).

Also it is easy to obtain a seven-state solution which does not satisfy Balzer's condition 3 (introduce $(GBG) \rightarrow R$ and $(GRG) \rightarrow F$ and $(GRX) \rightarrow F$, where R is a new state).

The remaining open question is: "What is the minimal number of states for a minimal time solution of the firing squad synchronization problem?". We know now that this number is 5 or 6.

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